MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

HIGH PERFORMANCE EMULATION CONTROL III MODULE, KEY CODE FWB

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic high performance emulation control III module.

1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 1.543 failures per million operating hours for class I or 3.509 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be FWB.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 112 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

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TABLE I. Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
A	Isolation	II	S	10	INF	10	INF	10	INF	MOhm
B1-B6	Continuity	III	X	0	1	0	1	0	1	Ohm
C	Voo stress	IV	S	---------Module shall pass all remaining tests---------						
D1	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D2	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D3-D5	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D6,D7	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D8	Ibd	V	S	0.0	3.0	0.0	3.0	0.0	3.0	mA
D9	Ibd	V	S	0.0	2.5	0.0	2.5	0.0	2.5	mA
D10	Ibd	V	S	0.0	0.4	0.0	0.4	0.0	0.4	mA
D11-D13	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D14	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D15,D16	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D17	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D18-D20	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D21	Ibd	V	S	0.0	2.1	0.0	2.1	0.0	2.1	mA
D22	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D23	Ibd	V	S	0.0	2.0	0.0	2.0	0.0	2.0	mA
D24	Ibd	V	S	0.0	3.0	0.0	3.0	0.0	3.0	mA
D25	Ibd	V	S	0.0	2.0	0.0	2.0	0.0	2.0	mA
D26	Ibd	V	S	0.0	1.2	0.0	1.2	0.0	1.2	mA
D27	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D28	Ibd	V	S	0.0	2.0	0.0	2.0	0.0	2.0	mA
D29-D33	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D34	Ibd	V	S	-0.25	0.346	-0.25	0.346	-0.25	0.346	mA
D35-D41	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D42,D43	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D44	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ∓ 5°C		Class II 4/ -55 ± 5°C 85 ∓ 5°C		Units
				Min	Max	Min	Max	Min	Max	
D45,D46	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D47	Ibd	V	S	0.0	0.1	0 0	0.1	0.0	0.1	mA
E1	Ios	VI	S	-100.0	-40.0	-100.0	-40.0	-100.0	-40.0	mA
E2,E3	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E4-E10	Ios	VI	S	-325.0	-100.0	-325.0	-100.0	-325.0	-100.0	mA
E11	Ios	VI	S	-326.0	-101.0	-326.0	-101.0	-326.0	-101.0	mA
E12-E17	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E18-E20	Ios	VI	S	-100.0	-40.0	-100.0	-40.0	-100.0	-40.0	mA
E21	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150 0	-60.0	mA
E22	Ios	VI	S	-100.0	-40.0	-100.0	-40.0	-100.0	-40.0	mA
E23-E27	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E28	Ios	VI	S	-100.0	-40.0	-100.0	-40.0	-100.0	-40.0	mA
E29-E35	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E36	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E37,E38	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E39	Ios	VI	S	-260.0	-30.0	-260.0	-30.0	-260.0	-30.0	mA
E40,E41	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E42	Ios	VI	S	-100.0	-40.0	-100.0	-40.0	-100 0	-40.0	mA
F1	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F2	Iih	VII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
F3-F5	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F6	Iih	VII	X	-1.4	-0.54	-1.4	-0.54	-1.4	-0.54	mA
F7,F8	Iih	VII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
F9	Iih	VII	X	-6.0	0.15	-6.0	0.15	-6.0	0.15	mA
F10	Iih	VII	X	-7.0	0.2	-7 0	0.2	-7.0	0.2	mA
F11	Iih	VII	X	-6.0	0.08	-6.0	0.08	-6.0	0.08	mA
F12-F14	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F15	Iih	VII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
F16,F17	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F18	Iih	VII	X	-1.2	0.04	-1 2	0.04	-1.2	0.04	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 + 5°C 4/		Class I 4/ 0 + 5°C 60 + 5°C		Class II 4/ -55 + 5°C 85 + 5°C		Units
				Min	Max	Min	Max	Min	Max	
F19-F21	Ith	VII	X	-0.25	0.025	-0.25	0.025	-0.25	0.025	mA
F22	Ith	VII	X	-1.1	0.07	-1 1	0.07	-1.1	0.07	mA
F23	Ith	VII	X	-0.25	0.025	-0.25	0.025	-0.25	0.025	mA
F24	Ith	VII	X	-0.5	0.05	-0.5	0.05	-0.5	0.05	mA
F25	Ith	VII	X	-0.75	0.075	-0.75	0.075	-0.75	0.075	mA
F26	Ith	VII	X	-0.5	0.05	-0.5	0.05	-0.5	0.05	mA
F27	Ith	VII	X	-1.05	0.065	-1.05	0.065	-1.05	0.065	mA
F28	Ith	VII	X	-0.25	0.025	-0.25	0.025	-0.25	0.025	mA
F29	Ith	VII	X	-0.5	0.05	-0.5	0.05	-0.5	0.05	mA
F30-F34	Ith	VII	X	-0.25	0.025	-0 25	0.025	-0.25	0.025	mA
F35	Ith	VII	X	-11.7	-6.1	-11.7	-6.1	-11.7	-6.1	mA
F36-F39	Ith	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F40	Ith	VII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
F41,F42	Ith	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F43,F44	Ith	VII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
F45	Ith	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F46,F47	Ith	VII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
F48	Ith	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G1	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G2	Iil	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G3-G5	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G6	Iil	VIII	X	-1.4	-0.54	-1.4	-0.54	-1.4	-0.54	mA
G7,G8	Iil	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G9	Iil	VIII	X	-6.0	0.15	-6.0	0.15	-6.0	0.15	mA
G10	Iil	VIII	X	-7.0	0.2	-7.0	0.2	-7.0	0.2	mA
G11	Iil	VIII	X	-6.0	0.08	-6 0	0.08	-6.0	0 08	mA
G12-G14	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G15	Iil	VIII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
G16,G17	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G18	Iil	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G19-G21	Iil	VIII	X	-0.25	0.025	-0 25	0.025	-0.25	0.025	mA

- See footnotes at end of table

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
G22	Iil	VIII	X	-1.1	0.07	-1 1	0.07	-1.1	0.07	mA
G23	Iil	VIII	X	-0.25	0.025	-0.25	0.025	-0.25	0.025	mA
G24	Iil	VIII	X	-0.5	0.05	-0.5	0.05	-0.5	0.05	mA
G25	Iil	VIII	X	-0.75	0.075	-0.75	0.075	-0.75	0.075	mA
G26	Iil	VIII	X	-0.5	0.05	-0.5	0.05	-0.5	0.05	mA
G27	Iil	VIII	X	-1.05	0.065	-1.05	0.065	-1.05	0.065	mA
G28	Iil	VIII	X	-0.25	0.025	-0.25	0.025	-0.25	0.025	mA
G29	Iil	VIII	X	-0.5	0.05	-0.5	0.05	-0.5	0.05	mA
G30-G34	Iil	VIII	X	-0.25	0.025	-0 25	0.025	-0.25	0.025	mA
G35	Iil	VIII	X	-11.7	-6.1	-11.7	-6.1	-11.7	-6.1	mA
G36-G39	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G40	Iil	VIII	X	-0.2	0.02	-0.2	0.02	-0.2	0.02	mA
G41,G42	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G43,G44	Iil	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G45	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G46,G47	Iil	VIII	X	-2.0	0.05	-2.0	0.05	-2.0	0.05	mA
G48	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
H1-H42	Voh	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J42	Vol	X	X	0.0	0.5	0.0	0.5	0.0	0.5	V
K1-K7	Iozh	XI	X	-0.05	0.05	-0.05	0.05	-0.05	0.05	mA
K8-K14	Iozl	XI	X	-0.05	0.05	-0.05	0.05	-0.05	0.05	mA
L1-L17	Vic	XII	S	-1.2	0.0	-1 2	0.0	-1.2	0.0	V
L18-L20	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L21	Vio	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L22-L25	Vic	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L26	Vio	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L27-L33	Vic	XII	S	-1.5	0.0	-1 5	0.0	-1.5	0.0	V
L34-L37	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
L38	Vio	XII	S	-1.5	0.0	-1.5	0.0	-1.5	0.0	V
L39-L46	Vic	XII	S	-1.2	0.0	-1 2	0.0	-1.2	0.0	V

- See footnotes at end of table.

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Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 80 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
M1	PSFC	XIII	S	3.7	6.9	3.7	6.9	3.7	6.9	uF
M2	PSFC	XIII	S	3.1	5.8	3.1	5.8	3.1	5.8	uF
N	Icc	XIV	X	400	1150	400	1150	400	1150	mA
P1,P2	Functional	XV	X	See table XV						
Q1,Q2	Dynamic	XVI	S	See table XVI						
Q3,Q4	Dynamic	XVI	Q	See table XVI						
R1	C1/o	XVIII	Q	2	75	2	75	2	75	pF
R2	C1/o	XVIII	Q	4	150	4	150	4	150	pF
R3	C1/o	XVIII	Q	2	75	2	75	2	75	pF
R4	C1/o	XVIII	Q	2	50	2	50	2	50	pF
R5	C1/o	XVIII	Q	2	75	2	75	2	75	pF
R6,R7	C1/o	XVIII	Q	4	75	4	75	4	75	pF
R8	C1/o	XVIII	Q	6	75	6	75	6	75	pF
R9	C1/o	XVIII	Q	10	175	10	175	10	175	pF
R10	C1/o	XVIII	Q	4	100	4	100	4	100	pF
R11	C1/o	XVIII	Q	2	50	2	50	2	50	pF
R12,R13	C1/o	XVIII	Q	2	75	2	75	2	75	pF
R14,R15	C1/o	XVIII	Q	2	50	2	50	2	50	pF
R16	C1/o	XVIII	Q	2	75	2	75	2	75	pF
R17	C1/o	XVIII	Q	4	75	4	75	4	75	pF
R18,R19	C1/o	XVIII	Q	2	50	2	50	2	50	pF
R20	C1/o	XVIII	Q	6	125	6	125	6	125	pF
R21	C1/o	XVIII	Q	2	75	2	75	2	75	pF
R22	C1/o	XVIII	Q	4	75	4	75	4	75	pF
R23	C1/o	XVIII	Q	6	100	6	100	6	100	pF
R24	C1/o	XVIII	Q	4	75	4	75	4	75	pF
R25	C1/o	XVIII	Q	6	125	6	125	6	125	pF
R26	C1/o	XVIII	Q	2	75	2	75	2	75	pF
R27	C1/o	XVIII	Q	4	100	4	100	4	100	pF

- See footnotes at end of table.

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Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
R28-R30	CI/O	XVIII	Q	2	50	2	50	2	50	pF
R31	CI/O	XVIII	Q	2	75	2	75	2	75	pF
R32	CI/O	XVIII	Q	2	100	2	100	2	100	pF
R33	CI/O	XVIII	Q	4	75	4	75	4	75	pF
R34-R36	CI/O	XVIII	Q	2	50	2	50	2	50	pF
R37,R38	CI/O	XVIII	Q	2	75	2	75	2	75	pF
R39,R40	CI/O	XVIII	Q	2	50	2	50	2	50	pF
R41	CI/O	XVIII	Q	4	75	4	75	4	75	pF
R42	CI/O	XVIII	Q	4	100	4	100	4	100	pF
R43	CI/O	XVIII	Q	2	50	2	50	2	50	pF
R44,R45	CI/O	XVIII	Q	2	125	2	125	2	125	pF
R46	CI/O	XVIII	Q	2	75	2	75	2	75	pF
R47	CI/O	XVIII	Q	6	125	6	125	6	125	pF
R48-R54	CI/O	XVIII	Q	2	75	2	75	2	75	pF
R55	CI/O	XVIII	Q	6	100	6	100	6	100	pF
R56	CI/O	XVIII	Q	2	50	2	50	2	50	pF
R57	CI/O	XVIII	Q	2	100	2	100	2	100	pF
R58	CI/O	XVIII	Q	6	100	6	100	6	100	pF
R59	CI/O	XVIII	Q	2	50	2	50	2	50	pF
T1,T2	PSTR	XIX	Q	0	400	0	400	0	400	mV

- See footnotes at end of table

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/542-1 for class I and M28787/542-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.5 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

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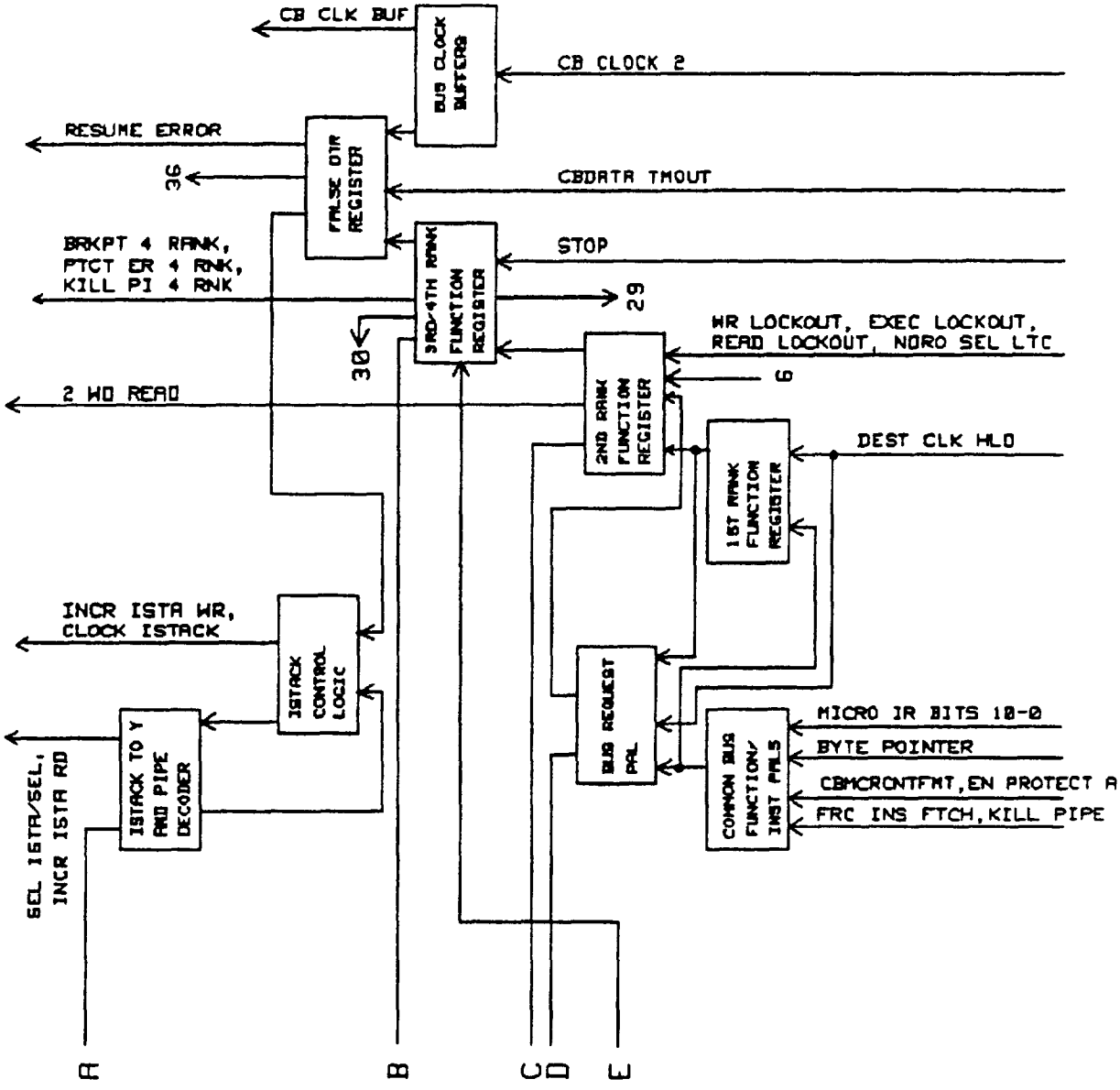


FIGURE 1. FWB functional block diagram.

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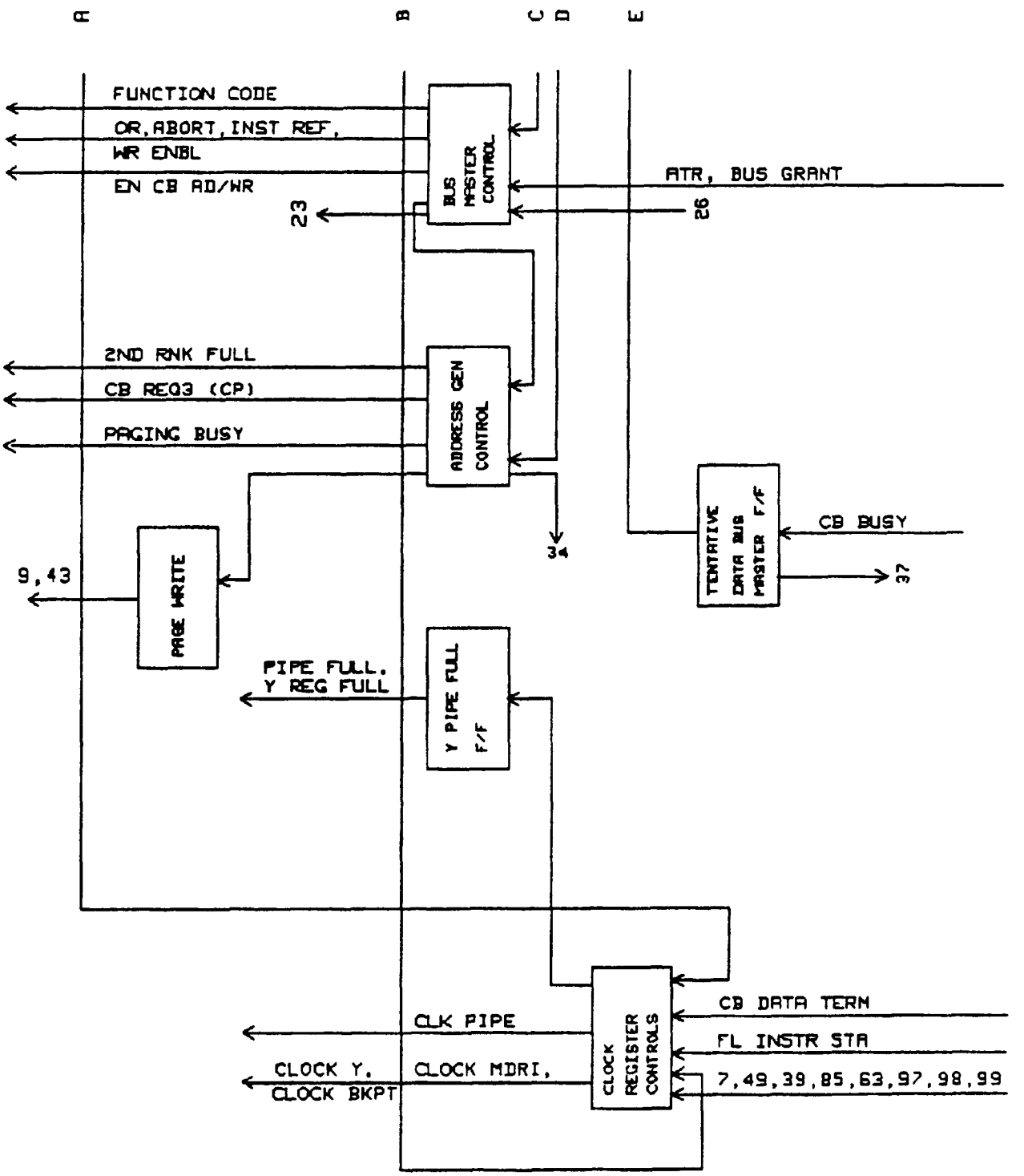


FIGURE 1. FWB functional block diagram - Continued.

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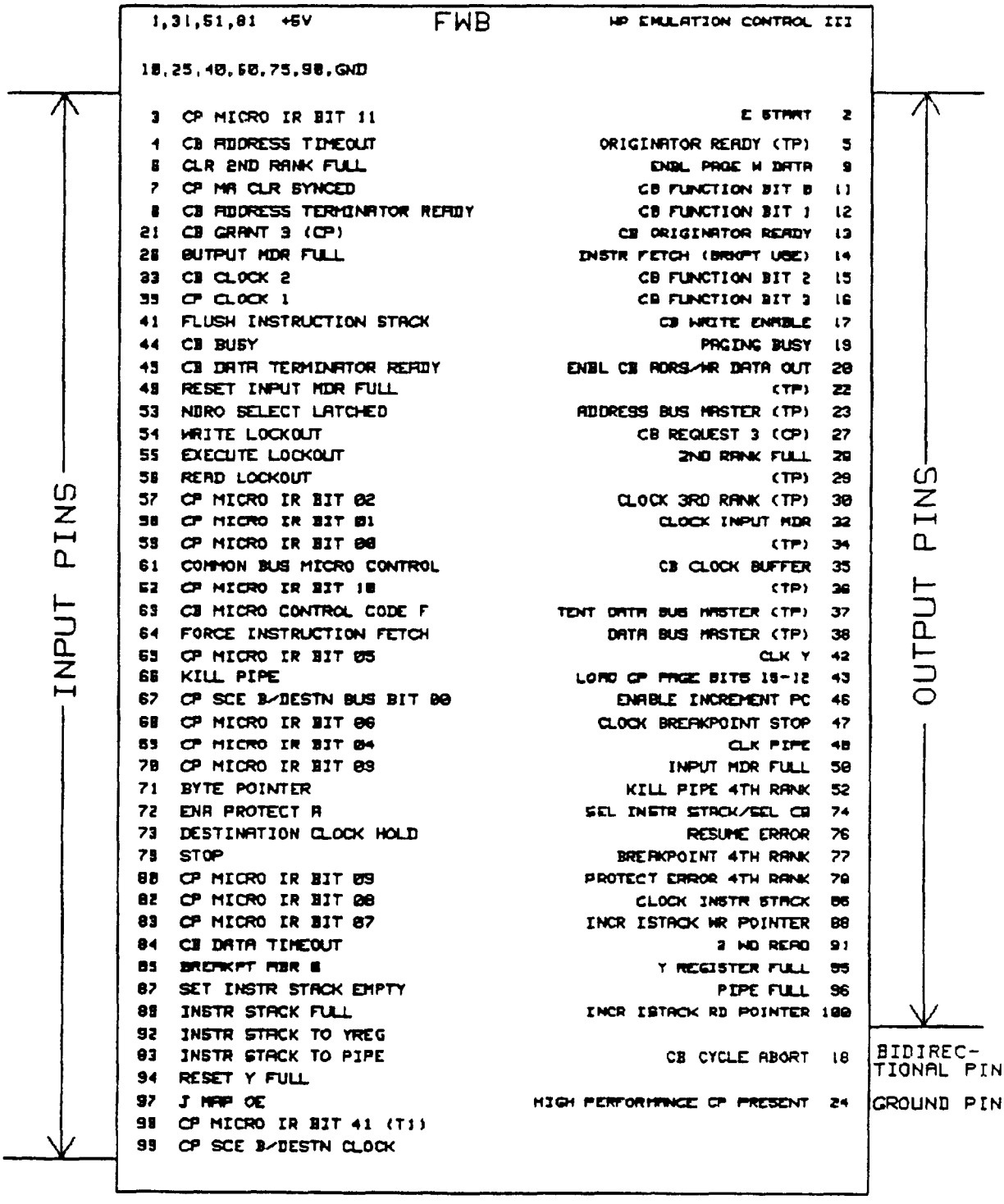


FIGURE 2. FWB pin assignment.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. Test conditions A, C, D, E, and L are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E or L are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, and L where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature

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stabilization period, test conditions B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition M, shall be performed using the initial limits at 25 ± 5°C ambient.

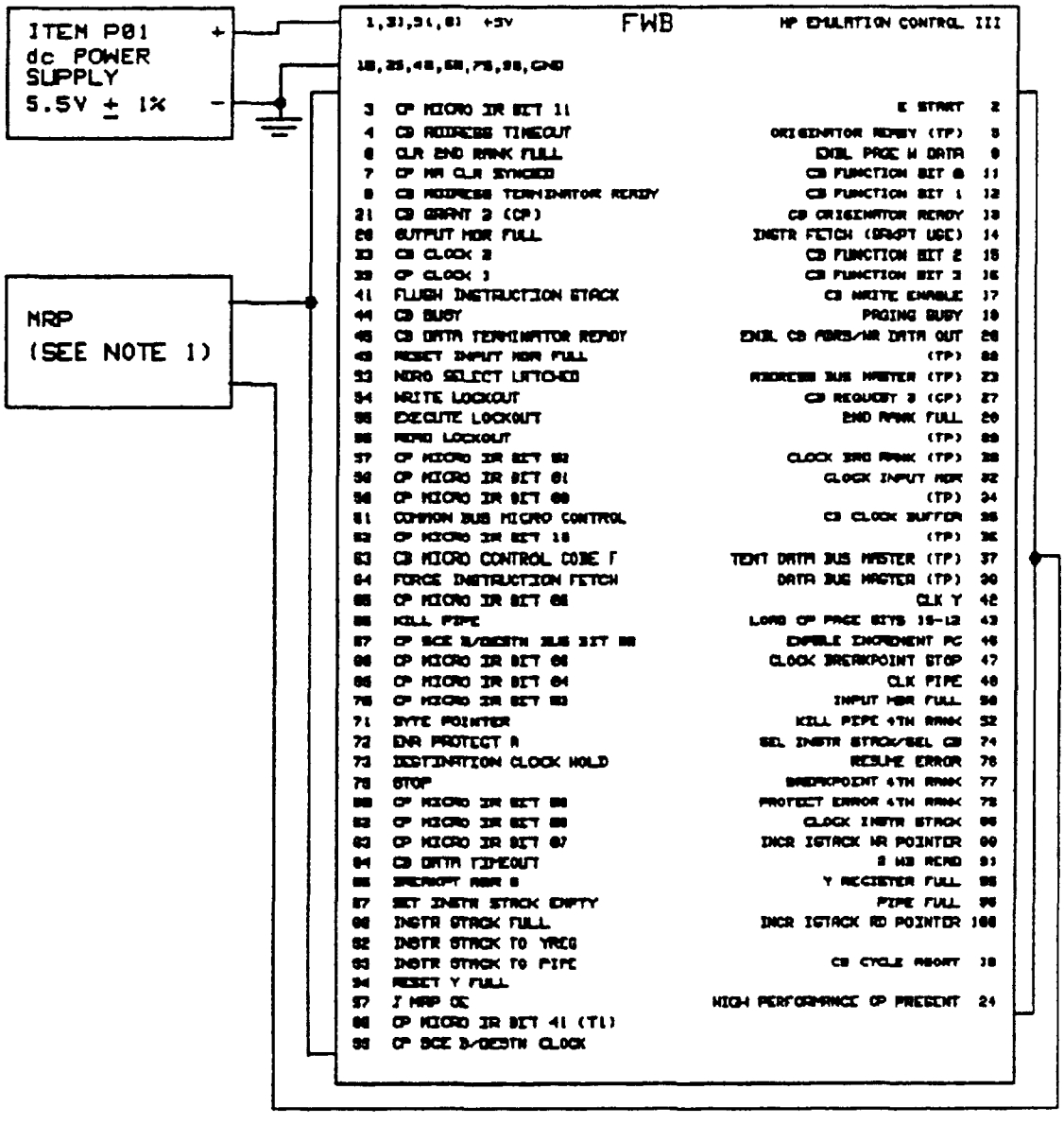
4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5. Following the life test and while still at the maximum class temperature, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4. Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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NOTE:

1. Military Reconfigurable Processor (MRP) or emulator capable of reproducing system backpanel characteristics.

FIGURE 3. FWB operating temperature and component temperature test circuit.

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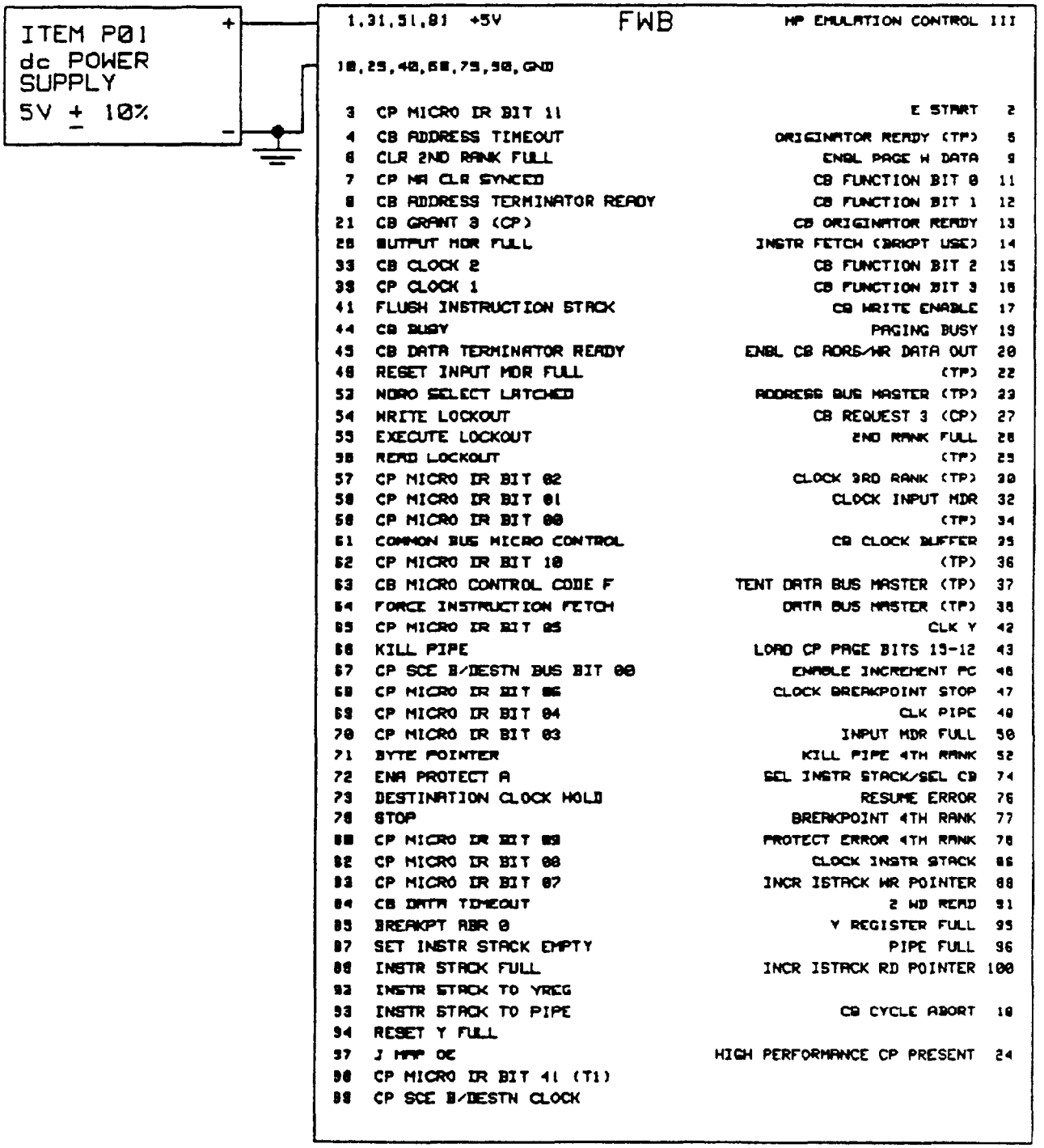
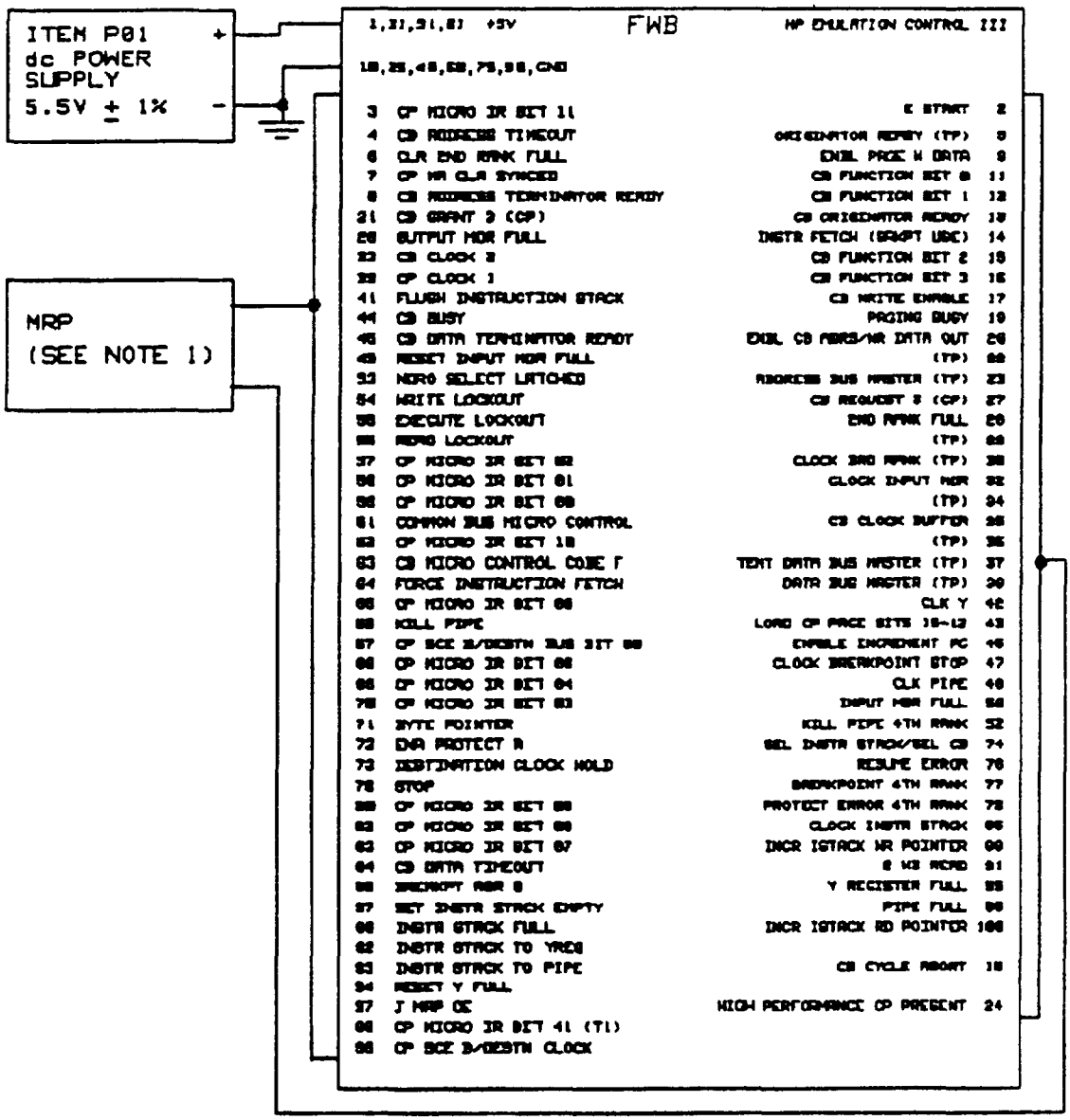


FIGURE 4. FWB vibration test circuit.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 5. FWB life test circuit.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ci/o = input/output capacitance.
(c) Cout = output capacitance.
(d) Ibd = input breakdown current.
(e) Icc = power supply current.
(f) Iih = input high level current.
(g) Iil = input low level current.
(h) Iin = input forcing current.
(i) Ioh = output high level current.
(j) Iol = output low level current.
(k) Ios = output short circuit current.
(l) Iozh = output high level leakage current.
(m) Iozl = output low level leakage current.
(n) NA = not applicable.
(o) PSFC = power supply filter capacitance.
(p) PSTA = power supply transient amplitude.
(q) Vic = input clamp diode voltage.
(r) Vih = input high level voltage.
(s) Vil = input low level voltage.
(t) Vin = input forcing voltage.
(u) Voh = output high level voltage.
(v) Vol = output low level voltage.
(w) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the -55°C to 85°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 6.325 W (max); 4.243 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
IR BIT 11	-1.2	0.4	2.4	-0.6	0.02	0.1	75
CB ADR TMOUT	-1.2	0.4	2.4	-1.2	0.04	0.2	150
CLR2 RNK FUL	-1.2	0.4	2.4	-0.6	0.02	0.1	75
CP MA CL SNC	-1.2	0.4	2.4	-0.6	0.02	0.1	50
CB ADDR TERM	-1.2	0.4	2.4	-0.6	0.02	0.1	75
CB GRANT 3	-1.2	0.4	2.4	-1.2	0.04	0.2	75
OTPT MDR FUL	-1.2	0.4	2.4	-1.2	0.04	0.2	75
CB CLOCK 2	-1.2	0.4	2.4	-6.0	0.15	3.0	75
CP CLOCK 1	-1.2	0.4	2.4	-7.0	0.2	2.5	175
FL INSTR STA	-1.2	0.4	2.4	-6.0	0.08	0.4	100
CB BUSY	-1.2	0.4	2.4	-0.6	0.02	0.1	50
CB DATA TERM	-1.2	0.4	2.4	-0.6	0.02	0.1	75
RESET IN MDR	-1.2	0.4	2.4	-0.6	0.02	0.1	75
NDRO SEL LAT	-1.2	0.4	2.4	-2.0	0.05	1.0	50

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
WR LOCKOUT	-1.2	0.4	2.4	-0.6	0.02	0.1	50
EXEC LOCKOUT	-1.2	0.4	2.4	-0.6	0.02	0.1	75
READ LOCKOUT	-1.2	0.4	2.4	-1.2	0.04	0.2	75
IR BIT 02	-1.5	0.4	2.4	-0.25	0.025	1.0	50
IR BIT 01	-1.5	0.4	2.4	-0.25	0.025	1.0	50
IR BIT 00	-1.5	0.4	2.4	-0.25	0.025	1.0	50
COM BUS CTRL	-1.2	0.4	2.4	-1.1	0.07	2.1	125
IR BIT 10	-1.5	0.4	2.4	-0.25	0.025	1.0	75
CB CTRL CD F	-1.5	0.4	2.4	-0.5	0.05	2.0	75
FRC INS FTCH	-1.5	0.4	2.4	-0.75	0.075	3.0	100
IR BIT 05	-1.5	0.4	2.4	-0.5	0.05	2.0	75
KILL PIPE	-1.2	0.4	2.4	-1.05	0.065	1.2	125
BUS BIT 00	-1.5	0.4	2.4	-0.25	0.025	1.0	75
IR BIT 06	-1.5	0.4	2.4	-0.5	0.05	2.0	100

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
IR BIT 04	-1.5	0.4	2.4	-0.25	0.025	1.0	50
IR BIT 03	-1.5	0.4	2.4	-0.25	0.025	1.0	50
BYTE POINTER	-1.5	0.4	2.4	-0.25	0.025	1.0	50
EN PROTECT A	-1.5	0.4	2.4	-0.25	0.025	1.0	75
DEST CLK HLD	-1.5	0.4	2.4	-0.25	0.025	1.0	100
STOP	NA	0.4	2.4	-11.7	-6.1	0.346	75
IR BIT 09	-1.2	0.4	2.4	-0.6	0.02	0.1	50
IR BIT 08	-1.2	0.4	2.4	-0.6	0.02	0.1	50
IR BIT 07	-1.2	0.4	2.4	-0.6	0.02	0.1	50
CBDATA TMOUT	-1.2	0.4	2.4	-0.6	0.02	0.1	75
BRKPT ABR 0	-1.5	0.4	2.4	-0.2	0.02	0.1	75
SET ISTA EMP	-1.2	0.4	2.4	-0.6	0.02	0.1	50
ISTACK FULL	-1.2	0.4	2.4	-0.6	0.02	0.1	50
ISTA TO YREG	-1.2	0.4	2.4	-1.2	0.04	0.2	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
ISTA TO PIPE	-1.2	0.4	2.4	-1.2	0.04	0.2	100
RESET Y FULL	-1.2	0.4	2.4	-0.6	0.02	0.1	50
J MAP OE	-1.2	0.4	2.4	-2.0	0.05	1.0	125
IR BIT 41	-1.2	0.4	2.4	-2.0	0.05	1.0	125
B/DESTN CLK	-1.2	0.4	2.4	-0.6	0.02	0.1	75
CB CYC ABORT	NA	0.4	2.4	-1.4	-0.54	NA	100

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
E START	2.4	0.5	-0.4	6.0	NA	NA	-40.0	NA
ORIG RDY TP	2.4	0.5	-0.4	6.0	NA	NA	-60.0	125
EN PG W DATA	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
CB FUNC BIT0	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	75
CB FUNC BIT1	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	75
ORIG RDY	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	75
STR FETCH	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	75
CB FUNC BIT2	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	75
CB FUNC BIT3	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	75
CB WRITE EN	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	75
PAGING BUSY	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
EN CB AD/WR	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100
(TP) 1	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
AD BUS MASTR	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
CB REQ3 (CP)	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
2ND RNK FULL	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
(TP) 2	2.4	0.5	-0.4	6.0	NA	NA	-40.0	NA
CLK 3RD RANK	2.4	0.5	-0.4	6.0	NA	NA	-40.0	NA
CLK IN MDR	2.4	0.5	-0.4	6.0	NA	NA	-40.0	NA
(TP) 3	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
CB CLK BUF	2.4	0.5	-0.4	6.0	NA	NA	-40.0	50
(TP) 4	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
TNT DT BS MR	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
DT BS MASTR	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
CLK Y	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
LD PG B15-12	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
EN INCR PC	2.4	0.5	-0.4	6.0	NA	NA	-40.0	100
CLK BRKPT ST	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
CLK PIPE	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
IN MDR FULL	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
KIL PI 4 RNK	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
SEL ISTA/SEL	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
RESUME ERROR	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
BRKPT 4 RANK	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
PTCT ER 4 RK	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
CLOCK ISTACK	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
INCR ISTA WR	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
2 WD READ	2.4	0.5	-0.16	1.6	NA	NA	-30.0	NA
Y REG FULL	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
PIPE FULL	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
INCR ISTA RD	2.4	0.5	-0.4	6.0	NA	NA	-40.0	NA
CB CYC ABORT	2.4	0.5	-0.4	6.0	NA	NA	-101.0	100

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6.3.3 Class I and II dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	CB CLOCK 2 to EN CB AD/WR	-	-	-	70	70	70	nS	12
1	tpd	CB CLOCK 2 to RESUME ERROR	-	-	-	70	70	70	nS	12
1	tpd	CB CLOCK 2 to PIPE FULL	-	-	-	70	70	70	nS	12
2	tpd	CB GRANT 3 to CB FUNC BIT0	-	-	-	50	50	50	nS	12
2	tpd	CB GRANT 3 to CB FUNC BIT1	-	-	-	50	50	50	nS	12
2	tpd	CB GRANT 3 to CB FUNC BIT2	-	-	-	50	50	50	nS	12
2	tpd	CB GRANT 3 to CB FUNC BIT3	-	-	-	50	50	50	nS	12
2	tpd	CB GRANT 3 to INSTR FETCH	-	-	-	50	50	50	nS	12
2	tpd	CB GRANT 3 to CB CYC ABORT	-	-	-	50	50	50	nS	12
3	tpd	B/DESTN CLK to PAGING BUSY	-	-	-	60	60	60	nS	12
4	tpd	CB CLOCK 2 to CB REQ3 (CP)	-	-	-	60	60	60	nS	12
4	tpd	CB CLOCK 2 to 2ND RNK FULL	-	-	-	60	60	60	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
4	tpd	CB CLOCK 2 to INCR ISTA WR	-	-	-	60	60	60	nS	12
4	tpd	CB CLOCK 2 to Y REG FULL	-	-	-	60	60	60	nS	12
5	tpd	CB CLOCK 2 to EN PG W DATA	-	-	-	40	40	40	nS	12
6	tpd	CB CLOCK 2 to CLK IN MDR	-	-	-	50	50	50	nS	12
6	tpd	CB CLOCK 2 to CLK Y	-	-	-	50	50	50	nS	12
6	tpd	CB CLOCK 2 to LD PG B15-12	-	-	-	50	50	50	nS	12
6	tpd	CB CLOCK 2 to CLK BRKPT ST	-	-	-	50	50	50	nS	12
6	tpd	CB CLOCK 2 to CLK PIPE	-	-	-	50	50	50	nS	12
7	tpd	CP CLOCK 1 to IN MDR FULL	-	-	-	100	100	100	nS	12
7	tpd	CP CLOCK 1 to SEL ISTA/SEL	-	-	-	100	100	100	nS	12
7	tpd	CP CLOCK 1 to 2 WD READ	-	-	-	100	100	100	nS	12
7	tpd	CP CLOCK 1 to INCR ISTA RD	-	-	-	100	100	100	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
8	tpd	CB CLOCK 2 to KIL PI 4 RNK	-	-	-	110	110	110	nS	12
8	tpd	CB CLOCK 2 to PTCT ER 4 RK	-	-	-	110	110	110	nS	12
8	tpd	CB CLOCK 2 to BRKPT 4 RANK	-	-	-	110	110	110	nS	12
9	tpd	ISTACK FULL to CLOCK ISTACK	-	-	-	20	20	20	nS	12
10	th	CB CLOCK 2 to CB ADDR TERM	50	50	50	-	-	-	nS	12
10	th	CB CLOCK 2 to CB GRANT 3	50	50	50	-	-	-	nS	12
11	tsu	CB ADR TMOUT to CB CLOCK 2	20	20	20	-	-	-	nS	12
11	tsu	OTPT MDR FUL to CB CLOCK 2	20	20	20	-	-	-	nS	12
12	th	CB CLOCK 2 to CB ABR TMOUT	60	60	60	-	-	-	nS	12
12	th	CB CLOCK 2 to OTPT MDR FUL	60	60	60	-	-	-	nS	12
13	tsu	CB BUSY to CB CLOCK 2	10	10	10	-	-	-	nS	12
13	tsu	CB DATA TERM to CB CLOCK 2	10	10	10	-	-	-	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
13	tsu	STOP to CB CLOCK 2	10	10	10	-	-	-	nS	12
13	tsu	CBDATA TMOUT to CB CLOCK 2	10	10	10	-	-	-	nS	12
13	tsu	BRKPT ABR 0 to CB CLOCK 2	10	10	10	-	-	-	nS	12
13	tsu	SET ISTA EMP to CB CLOCK 2	10	10	10	-	-	-	nS	12
14	th	CB CLOCK 2 to CB BUSY	50	50	50	-	-	-	nS	12
14	th	CB CLOCK 2 to CB DATA TERM	50	50	50	-	-	-	nS	12
14	th	CB CLOCK 2 to STOP	50	50	50	-	-	-	nS	12
14	th	CB CLOCK 2 to CBDATA TMOUT	50	50	50	-	-	-	nS	12
14	th	CB CLOCK 2 to BRKPT ABR 0	50	50	50	-	-	-	nS	12
14	th	CB CLOCK 2 to SET ISTA EMP	50	50	50	-	-	-	nS	12
15	tsu	RESET IN MDR to CP CLOCK 1	100	100	100	-	-	-	nS	12
15	tsu	ISTA TO YREG to CP CLOCK 1	100	100	100	-	-	-	nS	12
15	tsu	ISTA TO PIPE to CP CLOCK 1	100	100	100	-	-	-	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
16	th	CP CLOCK 1 to RESET IN MDR	20	20	20	-	-	-	nS	12
16	th	CP CLOCK 1 to ISTA TO YREG	20	20	20	-	-	-	nS	12
16	th	CP CLOCK 1 to ISTA TO PIPE	20	20	20	-	-	-	nS	12
17	tsu	RESET Y FULL to CP CLOCK 1	25	25	25	-	-	-	nS	12
18	th	CP CLOCK 1 to RESET Y FULL	15	15	15	-	-	-	nS	12
19	tsu	COM BUS CTRL to CP CLOCK 1	100	100	100	-	-	-	nS	12
19	tsu	CB CTRL CD F to CP CLOCK 1	100	100	100	-	-	-	nS	12
19	tsu	FRC INS FTCH to CP CLOCK 1	100	100	100	-	-	-	nS	12
19	tsu	KILL PIPE to CP CLOCK 1	100	100	100	-	-	-	nS	12
19	tsu	BUS BIT 00 CP CLOCK 1	100	100	100	-	-	-	nS	12
19	tsu	BYTE POINTER to CP CLOCK 1	100	100	100	-	-	-	nS	12
19	tsu	EN PROTECT A to CP CLOCK 1	100	100	100	-	-	-	nS	12

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6.3.3 Class I and II dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
19	tsu	DEST CLK HLD to CP CLOCK 1	100	100	100	-	-	-	nS	12
20	th	CP CLOCK 1 to COM BUS CTRL	20	20	20	-	-	-	nS	12
20	th	CP CLOCK 1 to CB CTRL CD F	20	20	20	-	-	-	nS	12
20	th	CP CLOCK 1 to FRC INS FTCH	20	20	20	-	-	-	nS	12
20	th	CP CLOCK 1 to KILL PIPE	20	20	20	-	-	-	nS	12
20	th	CP CLOCK 1 to BUS BIT 00	20	20	20	-	-	-	nS	12
20	th	CP CLOCK 1 to BYTE POINTER	20	20	20	-	-	-	nS	12
20	th	CP CLOCK 1 to EN PROTECT A	20	20	20	-	-	-	nS	12
20	th	CP CLOCK 1 to DEST CLK HLD	20	20	20	-	-	-	nS	12

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6.4 Module function. Emulation control provides the control logic to emulate the macroinstructions, process interrupts, communicate over the common bus, and master clear the MRP. High performance emulation control is contained on three modules; Emulation Control 1 (EWAH), 2 (FWA), and 3 (FWB). Macroinstructions are loaded into the pipe register from memory via the common bus. For double length instructions, the Y word is loaded into the Y register. Macroinstruction emulation is performed by executing a series of short microprogram subroutines stored in micro memory. The subroutine starting address is contained in an Emulation Control Word (ECW) stored in the ECW Programmable Read Only Memory (PROM). The ECW contains control bits along with the address pointer. A group of eight words are associated with each macroinstruction. The group is addressed by the macroinstruction Op and format codes (bits 15-8). An ECW within the group is addressed by address control. Address control determines if indexing, indirect addressing or both is specified, the instruction is being executed remotely, and the interim sequence required to execute the instruction. Status register 2 (bits 15-18) holds the indirect addressing information. Status register 1 holds Central Processor (CP) status bits that provide interrupt class lockouts, results of arithmetic operations, selection of page register sets, general register sets, Nondestructive Readout (NDRO) memory, and CP mode selection (executive mode or task mode). The executive mode has a separate set of privileged instructions that cannot be executed in the task mode (an attempt to execute a privileged instruction in the task mode results in a class II interrupt). Executive mode instructions are flagged. Conditional jump instructions (Op code = 40 (base 8)) are interpreted by conditional jump/stop control to determine if the specified jump condition is true. If true, the next macroinstruction is fetched from the jump address memory location. Interrupt control detects interrupts, determines the interrupt with the highest priority and generates a 4-bit interrupt pointer. The interrupt pointer is used to form the starting address of the microprogram subroutine that processes the interrupt. Bus master control provides the CP control interface with the common bus. The CP is always the bus master when communicating with the Input/Output Controller (IOC) or memory interfaces. The bus is requested from the bus arbitrator; and when the request is granted, bus master control generates a 4-bit function code and the control signals to perform the transaction. A MRP master clear is initiated when the MRP is powered-up or by the maintenance processor. Master clear places the MRP in the following initial condition:

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(a) P register cleared.

(b) Status register 1 and 2 cleared.

(c) Real Time Clock (RTC) and monitor clock functions disabled.

(d) Page registers are set to their own register address.

(e) All Input/Output (I/O) channels cleared as follows:

1. Status words cleared.
2. All data buffers and chains deactivated.
3. All interrupt data storage disabled, all External Interrupt Enables (EIE) cleared.
4. Class III interrupts disabled.
5. Serial monitor and suppress flags cleared.
6. Intercomputer time out function cleared.

(f) Bus initialization signal generated.

(g) Run mode selected (only occurs on power-up master clear).

(h) Program and power fault cleared.

6.4.1 Instruction Stack (ISTACK) controls. ISTACK controls are located on emulation control 2 and 3 modules. The following controls are located on the emulation control 3 module:

(a) ISTACK ready.

(b) Increment read/write pointer logic.

(c) Instruction multiplexer selects.

The ISTACK ready signal line is set when there is at least one word in the ISTACK that is available to read (that is, stable on the output lines). The ISTACK ready flip-flop is reset if the stack is empty, or it contains at least one word but the outputs are not yet stable. The ISTACK to Y and pipe decoder circuits select either the common bus or ISTACK input to the pipe register by controlling the instruction multiplexer. The ISTACK write pointer is incremented only if the memory reference is an ISTACK read fetch and the data is not loaded into the Y or pipe register. The ISTACK read pointer is incremented every time a

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word is transferred to the pipe or Y register from ISTACK. The instruction multiplexer normally selects the common bus read data. It selects the ISTACK if an ISTACK-to-pipe or ISTACK-to-Y transfer is performed. Additionally, the instruction multiplexer selects the ISTACK if the ISTACK is not empty and an instruction to Y or instruction to pipe microinstruction is performed.

6.4.2 Bus control logic. Bus control logic is composed of the bus master logic, tentative Data Bus Master (DBM) flip-flop, address generation control logic, the false Data Terminator Ready (DTR) register, the bus request Programmable Array Logic (PAL), and the clock register controls. Bus requests are sent to the bus arbitrator module as the result of a set bus request output from the bus request PAL. Bus grants from the bus arbitrator module are received at the bus master logic which generate enables for function bits, the OR signal, and initiates a tentative DBM condition. Address Terminator Ready (ATR) and busy signals are generated by the bus slave module. Address and data timeouts cause a resume error to be generated by the macro/micro interrupt logic on the high performance emulation control 2 module. Additionally, address timeout and data timeout signals are used to generate a false internal DTR which is used to reset the bus control logic. A received DTR at the clock register controls also resets the bus control logic.

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6.4.3 Description of I/O pins.

Signal name	Pin number	I/O	Description
IR BIT 11	3	I	CP MICRO IR BIT 11
CB ADR TMOUT	4	I	CB ADDRESS TIMEOUT
CLR2 RNK FUL	6	I	CLR 2ND RANK FULL
CP MA CL SNC	7	I	CP MA CLR SYNCED
CB ADDR TERM	8	I	CB ADDR TERMINATOR READY
CB GRANT 3	21	I	CB GRANT 3 (CP)
OTPT MDR FUL	26	I	OUTPUT MDR FULL
CB CLOCK 2	33	I	CB CLOCK 2
CP CLOCK 1	39	I	CP CLOCK 1
FL INSTR STA	41	I	FLUSH INSTRUCTION STACK
CB BUSY	44	I	CB BUSY
CB DATA TERM	45	I	CB DATA TERMINATOR READY
RESET IN MDR	49	I	RESET INPUT MDR FULL
NDRO SEL LAT	53	I	NDRO SELECT LATCHED
WR LOCKOUT	54	I	WRITE LOCKOUT
EXEC LOCKOUT	55	I	EXECUTE LOCKOUT
READ LOCKOUT	56	I	READ LOCKOUT
IR BIT 02	57	I	CP MICRO IR BIT 02
IR BIT 01	58	I	CP MICRO IR BIT 01
IR BIT 00	59	I	CP MICRO IR BIT 00

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6.4.3 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
COM BUS CTRL	61	I	COMMON BUS MICRO CONTROL
IR BIT 10	62	I	CP MICRO IR BIT 10
CB CTRL CD F	63	I	CB CONTROL CODE F
FRC INS FTCH	64	I	FORCE INSTRUCTION FETCH
IR BIT 05	65	I	MICRO IR BIT 05
KILL PIPE	66	I	KILL PIPE
BUS BIT 00	67	I	CP SCE B/DESTN BUS BIT 00
IR BIT 06	68	I	CP MICRO IR BIT 06
IR BIT 04	69	I	CP MICRO IR BIT 04
IR BIT 03	70	I	CP MICRO IR BIT 03
BYTE POINTER	71	I	BYTE POINTER
EN PROTECT A	72	I	ENA PROTECT A
DEST CLK HLD	73	I	DESTINATION CLOCK HOLD
STOP	79	I	STOP
IR BIT 09	80	I	CP MICRO IR BIT 09
IR BIT 08	82	I	CP MICRO IR BIT 08
IR BIT 07	83	I	CP MICRO IR BIT 07
CBDATA TMOUT	84	I	CB DATA TIMEOUT
BRKPT ABR 0	85	I	BREAKPT ABR 0
SET ISTA EMP	87	I	SET INSTR STACK EMPTY

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6.4.3 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
ISTACK FULL	89	I	INSTR STACK FULL
ISTA TO YREG	92	I	INSTR STACK TO Y REG
ISTA TO PIPE	93	I	INSTR STACK TO PIPE
RESET Y FULL	94	I	RESET Y FULL
J MAP OE	97	I	J MAP OE
IR BIT 41	98	I	CP MICRO IR BIT 41 (T1)
B/DESTN CLK	99	I	CP SCE B/DESTN CLOCK
E START	2	O	E START
ORIG RDY TP	5	O	ORIGINATOR READY (TP)
EN PG W DATA	9	O	ENBL PAGE W DATA
CB FUNC BIT0	11	O	CB FUNCTION BIT 0
CB FUNC BIT1	12	O	CB FUNCTION BIT 1
CB ORIG RDY	13	O	CB ORGINATOR READY
INSTR FETCH	14	O	INSTR FETCH (BRKPT USE)
CB FUNC BIT2	15	O	CB FUNCTION BIT 2
CB FUNC BIT3	16	O	CB FUNCTION BIT 3
CB WRITE EN	17	O	CB WRITE ENABLE
PAGING BUSY	19	O	PAGING BUSY
EN CB AD/WR	20	O	ENBL CB ADRS/WR DATA OUT
(TP) 1	22	O	(TP) 1

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6.4.3 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
AD BUS MASTR	23	O	ADDRESS BUS MASTER (TP)
CB REQ3 (CP)	27	O	CB REQUEST 3 (CP)
2ND RNK FULL	28	O	2ND RANK FULL
(TP) 2	29	O	(TP) 2
CLK 3RD RANK	30	O	CLOCK 3RD RANK (TP)
CLK IN MDR	32	O	CLOCK INPUT MDR
(TP) 3	34	O	(TP) 3
CB CLK BUF	35	O	CB CLOCK BUFFER
(TP) 4	36	O	(TP) 4
TNT DT BS MR	37	O	TENT DATA BUS MASTER (TP)
DT BS MASTR	38	O	DATA BUS MASTER (TP)
CLK Y	42	O	CLK Y
LD PG B15-12	43	O	LOAD CP PAGE BITS 15-12
EN INCR PC	46	O	ENABLE INCREMENT PC
CLK BRKPT ST	47	O	CLOCK BREAKPOINT STOP
CLK PIPE	48	O	CLK PIPE
IN MDR FULL	50	O	INPUT MDR FULL
KIL PI 4 RNK	52	O	KILL PIPE 4TH RANK
SEL ISTA/SEL	74	O	SEL INSTR STACK/SEL CB
RESUME ERROR	76	O	RESUME ERROR

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6.4.3 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
BRKPT 4 RANK	77	O	BREAKPOINT 4TH RANK
PTCT ER 4 RK	78	O	PROTECT ERROR 4TH RANK
CLOCK ISTACK	86	O	CLOCK INSTR STACK
INCR ISTA WR	88	O	INCREMENT ISTACK WRITE POINTER
2 WD READ	91	O	2 WD READ
Y REG FULL	95	O	Y REGISTER FULL
PIPE FULL	96	O	PIPE FULL
INCR ISTA RD	100	O	INCREMENT ISTACK READ POINTER
CB CYC ABORT	18	I/O	CB CYCLE ABORT

6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Preparing activity:
Navy - SH

(Project 5963-N718)

Agent:
NWSC Crane

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TABLE II. Isolation test.

Test cond	Test description	Connect		25 ± 5°C Initial limits		Units	Notes
		A to	B to	Min	Max		
A	Isolation, frame	C	Frame	10	INF	MOhm	1-8

NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A
Isolation measurement	NA	± 500 KOhm	A

2. The ohmmeter used shall be item R06.

3. In the set-up conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, and 'B' shall correspond to the negative terminal and 'C' shall correspond to all used pins shorted together.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

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NOTES: - Continued.

6. Caution: Set megohm bridge to limit current to 10 uA before applying voltage to the module.

7. The test circuit shall be as indicated on figure 6.

8. Abbreviations used are defined in 6.1.

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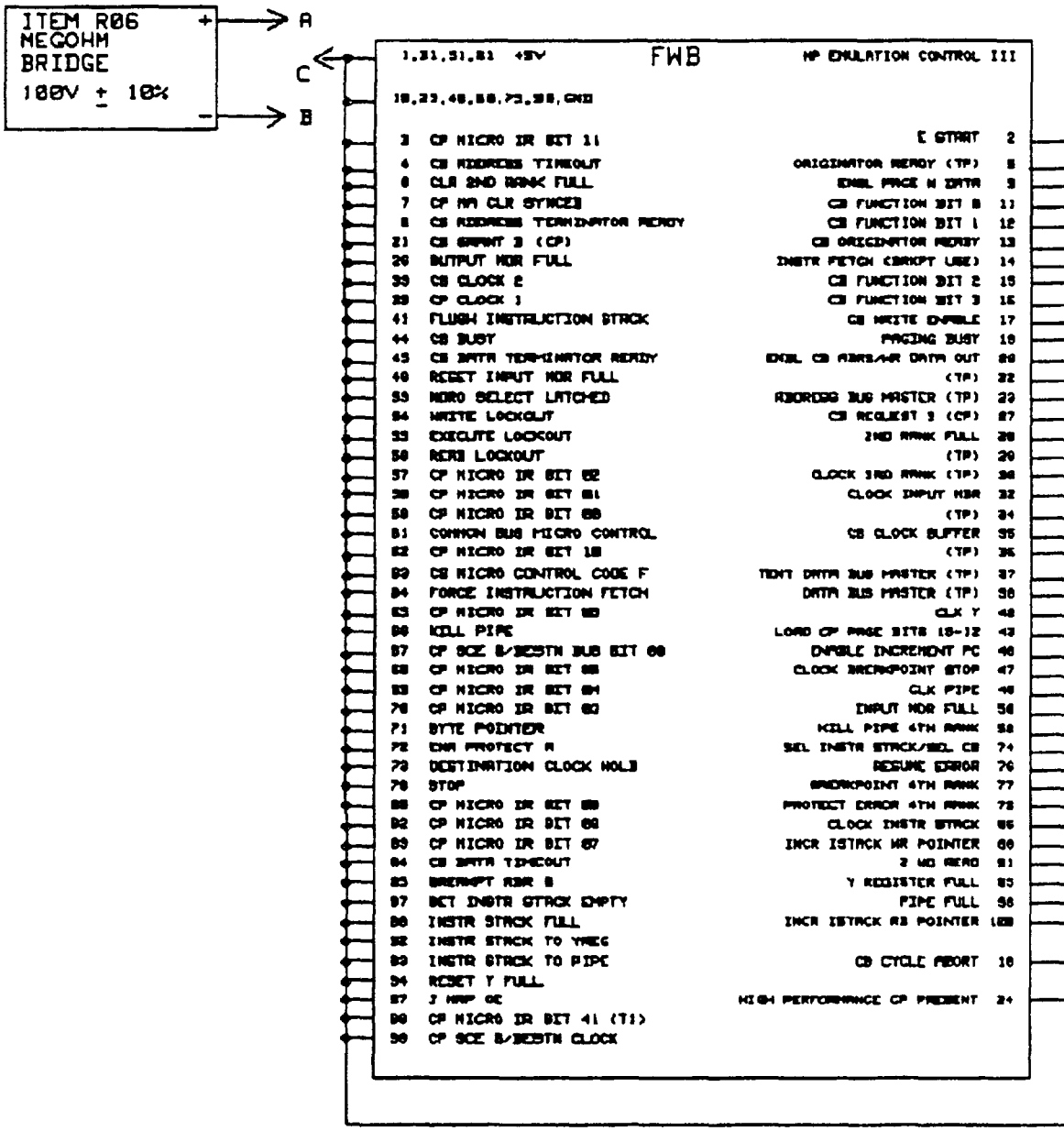


FIGURE 6. Isolation test circuit for table II.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to	Connect B to	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity, ground	10	25	0	1	Ohm	1-5
B2	Continuity, ground	10	40				
B3	Continuity, ground	60	75				
B4	Continuity, ground	60	90				
B5	Continuity, +5V	1	31				
B6	Continuity, +5V	51	81	0	1	Ohm	1-5

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA	± 0.05 Ohm	B1-B6

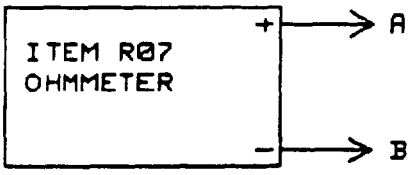
2. The ohmmeter used shall be item R07.

3. In the set-up conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. The test circuit shall be as indicated on figure 7.

5. Abbreviations used are defined in 6.1.

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1,31,51,81 +5V	FWB	MP EMULATION CONTROL III	
10,25,40,60,75,90,GND			
3 CP MICRO IR BIT 11		I START	2
4 CB ADDRESS TIMEOUT		ORIGINATOR READY (TP)	5
6 CLR 2ND RANK FULL		ENBL PAGE M DATA	9
7 CP MA CLR SYNCED		CB FUNCTION BIT 0	11
8 CB ADDRESS TERMINATOR READY		CB FUNCTION BIT 1	12
21 CB GRANT 3 (CP)		CB ORIGINATOR READY	13
28 OUTPUT MDR FULL		INSTR FETCH (BRKPT USE)	14
32 CB CLOCK 2		CB FUNCTION BIT 2	15
38 CP CLOCK 1		CB FUNCTION BIT 3	16
41 FLUSH INSTRUCTION STACK		CB WRITE ENABLE	17
44 CB BUSY		PAGING BUSY	19
48 CB DATA TERMINATOR READY		ENBL CB ADRS/WR DATA OUT	20
49 RESET INPUT MDR FULL		(TP)	22
53 MDRO SELECT LATCHED		ADDRESS BUS MASTER (TP)	23
54 WRITE LOCKOUT		CB REQUEST 3 (CP)	27
55 EXECUTE LOCKOUT		2ND RANK FULL	28
56 READ LOCKOUT		(TP)	29
57 CP MICRO IR BIT 02		CLOCK 3RD RANK (TP)	30
58 CP MICRO IR BIT 01		CLOCK INPUT MDR	32
59 CP MICRO IR BIT 00		(TP)	34
61 COMMON BUS MICRO CONTROL		CB CLOCK BUFFER	35
62 CP MICRO IR BIT 10		(TP)	36
63 CB MICRO CONTROL CODE F		TEXT DATA BUS MASTER (TP)	37
64 FORCE INSTRUCTION FETCH		DATA BUS MASTER (TP)	38
65 CP MICRO IR BIT 05		CLK Y	42
66 KILL PIPE		LOAD CP PAGE BITS 15-12	43
67 CP SCE B/DESTN BUS BIT 00		ENABLE INCREMENT PC	45
68 CP MICRO IR BIT 06		CLOCK BREAKPOINT STOP	47
69 CP MICRO IR BIT 04		CLK PIPE	48
70 CP MICRO IR BIT 03		INPUT MDR FULL	50
71 BYTE POINTER		KILL PIPE 4TH RANK	52
72 ENA PROTECT M		SEL INSTR STACK/SEL CB	74
73 DESTINATION CLOCK HOLD		RESUME ERROR	76
75 STOP		BREAKPOINT 4TH RANK	77
80 CP MICRO IR BIT 09		PROTECT ERROR 4TH RANK	78
82 CP MICRO IR BIT 08		CLOCK INSTR STACK	86
83 CP MICRO IR BIT 07		INCR ISTACK WR POINTER	88
84 CB DATA TIMEOUT		2 WD READ	91
85 BREAKPT ABR 0		Y REGISTER FULL	93
87 SET INSTR STACK EMPTY		PIPE FULL	96
89 INSTR STACK FULL		INCR ISTACK RD POINTER	100
92 INSTR STACK TO YREG			
93 INSTR STACK TO PIPE		CB CYCLE ABORT	19
94 RESET Y FULL			
97 J MAP OE		HIGH PERFORMANCE CP PRESENT	24
98 CP MICRO IR BIT 41 (T1)			
99 CP SCE B/DESTN CLOCK			

FIGURE 7. Continuity test circuit for table III.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect		25 ± 5°C Initial limits	Notes
			Positive lead to	Negative lead to		
C	Vcc	7.0	1,31,51, 81	10,25, 40,60, 75,90	Module shall pass all remaining tests.	1-6

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV, -70 mV	C
Power supply current limit	2.07 A	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. The test circuit shall be as indicated on figure 8.

6. Abbreviations used are defined in 6.1.

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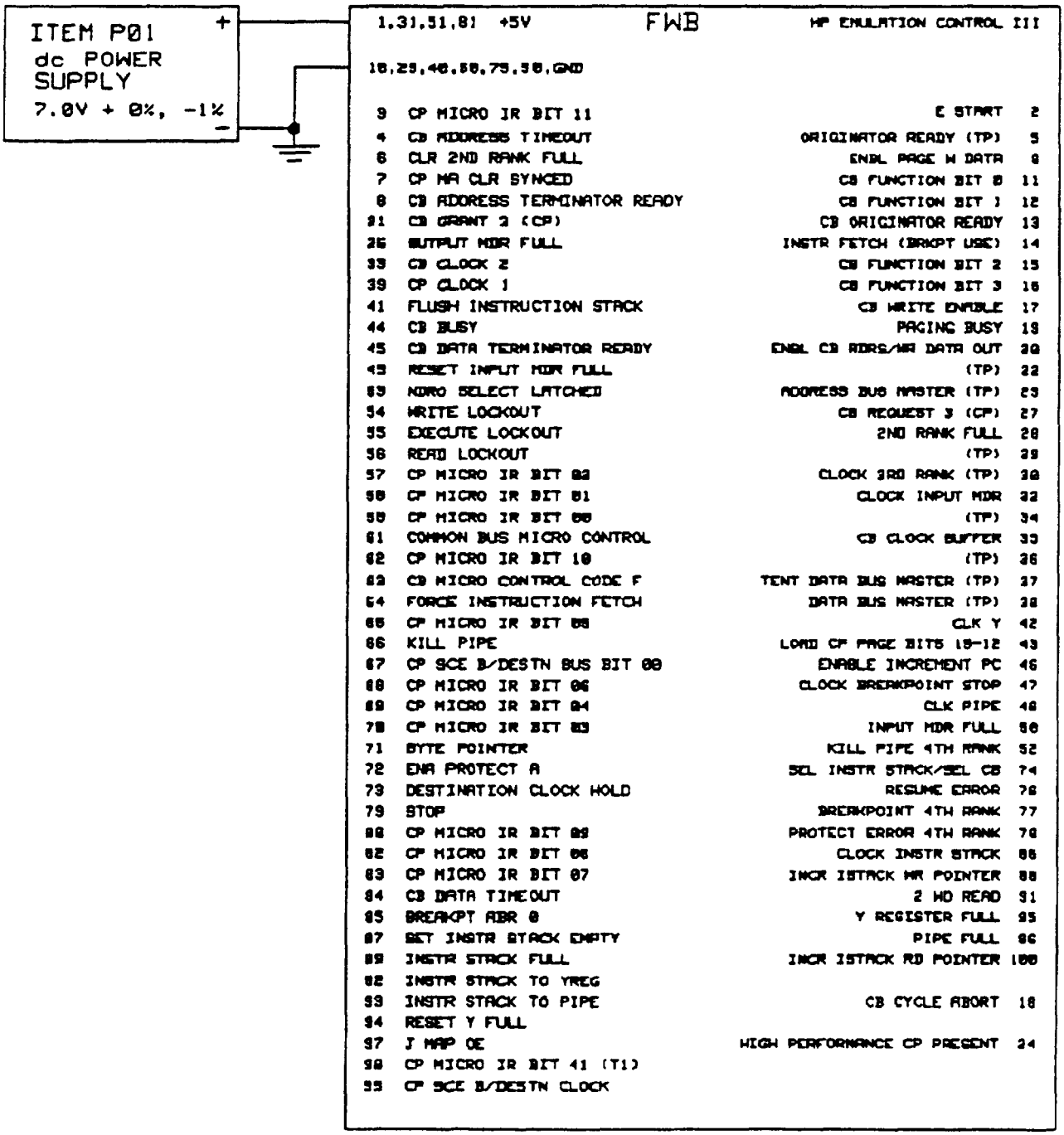


FIGURE 8. Vcc stress test circuit for table IV.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	IR BIT 11	3	5.5	0.0	0.1	mA	1-3
D2	CB ADR TMOUT	4	5.5	0.0	0.2	mA	1-3
D3	CLR2 RNK FUL	6	5.5	0.0	0.1	mA	1-3
D4	CP MA CL SNC	7					
D5	CB ADDR TERM	8	5.5	0.0	0.1	mA	1-3
D6	CB GRANT 3	21	5.5	0.0	0.2	mA	1-3
D7	OTPT MDR FUL	26	5.5	0.0	0.2	mA	1-3
D8	CB CLOCK 2	33	5.5	0.0	3.0	mA	1-3
D9	CP CLOCK 1	39	5.5	0.0	2.5	mA	1-3
D10	FL INSTR STA	41	5.5	0.0	0.4	mA	1-3
D11	CB BUSY	44	5.5	0.0	0.1	mA	1-3
D12	CB DATA TERM	45					
D13	RESET IN MDR	49	5.5	0.0	0.1	mA	1-3
D14	NDRO SEL LAT	53	5.5	0.0	1.0	mA	1-3
D15	WR LOCKOUT	54	5.5	0.0	0.1	mA	1-3
D16	EXEC LOCKOUT	55	5.5	0.0	0.1	mA	1-3
D17	READ LOCKOUT	56	5.5	0.0	0.2	mA	1-3
D18	IR BIT 02	57	5.5	0.0	1.0	mA	1-3
D19	IR BIT 01	58					
D20	IR BIT 00	59	5.5	0.0	1.0	mA	1-3
D21	COM BUS CTRL	61	5.5	0.0	2.1	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D22	IR BIT 10	62	5.5	0.0	1.0	mA	1-3
D23	CB CTRL CD F	63	5.5	0.0	2.0	mA	1-3
D24	FRC INS FTCH	64	5.5	0.0	3.0	mA	1-3
D25	IR BIT 05	65	5.5	0.0	2.0	mA	1-3
D26	KILL PIPE	66	5.5	0.0	1.2	mA	1-3
D27	BUS BIT 00	67	5.5	0.0	1.0	mA	1-3
D28	IR BIT 06	68	5.5	0.0	2.0	mA	1-3
D29	IR BIT 04	69	5.5	0.0	1.0	mA	1-3
D30	IR BIT 03	70					
D31	BYTE POINTER	71					
D32	EN PROTECT A	72					
D33	DEST CLK HLD	73	5.5	0.0	1.0	mA	1-3
D34	STOP	79	5.5	-0.25	0.346	mA	1-3
D35	IR BIT 09	80	5.5	0.0	0.1	mA	1-3
D36	IR BIT 08	82					
D37	IR BIT 07	83					
D38	CBDATA TMOUT	84					
D39	BRKPT ABR 0	85					
D40	SET ISTA EMP	87					
D41	ISTACK FULL	89	5.5	0.0	0.1	mA	1-3
D42	ISTA TO YREG	92	5.5	0.0	0.2	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D43	ISTA TO PIPE	93	5.5	0.0	0.2	mA	1-3
D44	RESET Y FULL	94	5.5	0.0	0.1	mA	1-3
D45	J MAP OE	97	5.5	0.0	1.0	mA	1-3
D46	IR BIT 41	98	5.5	0.0	1.0	mA	1-3
D47	B/DESTN CLK	99	5.5	0.0	0.1	mA	1-3

NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D47
Power supply current limit	1.73 A	NA	D1-D47
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D47
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 D1-D47 D1-D47
Input current measurement	NA NA NA NA NA	± 5 uA ± 10 uA ± 5 uA ± 10 uA ± 150 uA	D1 D2 D3-D5 D6,D7 D8

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA	± 125 uA	D9
	NA	± 20 uA	D10
	NA	± 5 uA	D11-D13
	NA	± 50 uA	D14
	NA	± 5 uA	D15,D16
	NA	± 10 uA	D17
	NA	± 50 uA	D18-D20
	NA	± 105 uA	D21
	NA	± 50 uA	D22
	NA	± 100 uA	D23
	NA	± 150 uA	D24
	NA	± 100 uA	D25
	NA	± 60 uA	D26
	NA	± 50 uA	D27
	NA	± 100 uA	D28
	NA	± 50 uA	D29-D33
	NA	± 30 uA	D34
	NA	± 5 uA	D35-D41
	NA	± 10 uA	D42,D43
	NA	± 5 uA	D44
	NA	± 50 uA	D45,D46
	NA	± 5 uA	D47

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

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NOTES: - Continued.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	E START	2	0.0	-100.0	-40.0	mA	1-4
E2	ORIG RDY TP	5	0.0	-150.0	-60.0	mA	1-4
E3	EN PG W DATA	9	0.0	-150.0	-60.0	mA	1-4
E4	CB FUNC BIT0	11	0.0	-325.0	-100.0	mA	1-4
E5	CB FUNC BIT1	12					
E6	CB ORIG RDY	13					
E7	INSTR FETCH	14					
E8	CB FUNC BIT2	15					
E9	CB FUNC BIT3	16					
E10	CB WRITE EN	17	0.0	-325.0	-100.0	mA	1-4
E11	CB CYC ABORT	18	0.0	-326.0	-101.0	mA	1-4
E12	PAGING BUSY	19	0.0	-150.0	-60.0	mA	1-4
E13	EN CB AD/WR	20					
E14	(TP) 1	22					
E15	AD BUS MASTR	23					
E16	CB REQ3 (CP)	27					
E17	2ND RNK FULL	28	0.0	-150.0	-60.0	mA	1-4
E18	(TP) 2	29	0.0	-100.0	-40.0	mA	1-4
E19	CLK 3RD RANK	30					
E20	CLK IN MDR	32	0.0	-100.0	-40.0	mA	1-4
E21	(TP) 3	34	0.0	-150.0	-60.0	mA	1-4

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Check the source to verify that this is the current version before use.

TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E22	CB CLK BUF	35	0.0	-100.0	-40.0	mA	1-4
E23	(TP) 4	36	0.0	-150.0	-60.0	mA	1-4
E24	TNT DT BS MR	37					
E25	DT BS MASTR	38					
E26	CLK Y	42					
E27	LD PG B15-12	43	0.0	-150.0	-60.0	mA	1-4
E28	EN INCR PC	46	0.0	-100.0	-40.0	mA	1-4
E29	CLK BRKPT ST	47	0.0	-150.0	-60.0	mA	1-4
E30	CLK PIPE	48					
E31	IN MDR FULL	50					
E32	KIL PI 4 RNK	52					
E33	SEL ISTA/SEL	74					
E34	RESUME ERROR	76					
E35	BRKPT 4 RANK	77	0.0	-150.0	-60.0	mA	1-4
E36	PTCT ER 4 RK	78	0.0	-260.0	-30.0	mA	1-4
E37	CLOCK ISTACK	86	0.0	-150.0	-60.0	mA	1-4
E38	INCR ISTA WR	88	0.0	-150.0	-60.0	mA	1-4
E39	2 WD READ	91	0.0	-260.0	-30.0	mA	1-4
E40	Y REG FULL	95	0.0	-150.0	-60.0	mA	1-4
E41	PIPE FULL	96	0.0	-150.0	-60.0	mA	1-4
E42	INCR ISTA RD	100	0.0	-100.0	-40.0	mA	1-4

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NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	E1-E42
Power supply current limit	1.73 A	NA	E1-E42
Forcing voltage, (Vout)	0.0 V	± 50 mV	E1-E42
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 E1-E42 E1-E42
Output current measurement	NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA	± 3.00 mA ± 4.50 mA ± 6.25 mA ± 4.50 mA ± 3.00 mA ± 4.50 mA ± 3.00 mA ± 4.50 mA ± 3.00 mA ± 4.50 mA ± 5.50 mA ± 4.50 mA ± 5.50 mA ± 4.50 mA ± 3.00 mA	E1 E2,E3 E4-E11 E12-E17 E18-E20 E21 E22 E23-E27 E28 E29-E35 E36 E37,E38 E39 E40,E41 E42

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	IR BIT 11	3	2.4	-0.6	0.02	mA	1-3
F2	CB ADR TMOUT	4	2.4	-1.2	0.04	mA	1-3
F3	CLR2 RNK FUL	6	2.4	-0.6	0.02	mA	1-3
F4	CP MA CL SNC	7					
F5	CB ADDR TERM	8	2.4	-0.6	0.02	mA	1-3
F6	CB CYC ABORT	18	2.4	-1.4	-0.54	mA	1-3
F7	CB GRANT 3	21	2.4	-1.2	0.04	mA	1-3
F8	OTPT MDR FUL	26	2.4	-1.2	0.04	mA	1-3
F9	CB CLOCK 2	33	2.4	-6.0	0.15	mA	1-3
F10	CP CLOCK 1	39	2.4	-7.0	0.2	mA	1-3
F11	FL INSTR STA	41	2.4	-6.0	0.08	mA	1-3
F12	CB BUSY	44	2.4	-0.6	0.02	mA	1-3
F13	CB DATA TERM	45					
F14	RESET IN MDR	49	2.4	-0.6	0.02	mA	1-3
F15	NDRO SEL LAT	53	2.4	-2.0	0.05	mA	1-3
F16	WR LOCKOUT	54	2.4	-0.6	0.02	mA	1-3
F17	EXEC LOCKOUT	55	2.4	-0.6	0.02	mA	1-3
F18	READ LOCKOUT	56	2.4	-1.2	0.04	mA	1-3
F19	IR BIT 02	57	2.4	-0.25	0.025	mA	1-3
F20	IR BIT 01	58					
F21	IR BIT 00	59	2.4	-0.25	0.025	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F22	COM BUS CTRL	61	2.4	-1.1	0.07	mA	1-3
F23	IR BIT 10	62	2.4	-0.25	0.025	mA	1-3
F24	CB CTRL CD F	63	2.4	-0.5	0.05	mA	1-3
F25	FRC INS FTCH	64	2.4	-0.75	0.075	mA	1-3
F26	IR BIT 05	65	2.4	-0.5	0.05	mA	1-3
F27	KILL PIPE	66	2.4	-1.05	0.065	mA	1-3
F28	BUS BIT 00	67	2.4	-0.25	0.025	mA	1-3
F29	IR BIT 06	68	2.4	-0.5	0.05	mA	1-3
F30	IR BIT 04	69	2.4	-0.25	0.025	mA	1-3
F31	IR BIT 03	70					
F32	BYTE POINTER	71					
F33	EN PROTECT A	72					
F34	DEST CLK HLD	73	2.4	-0.25	0.025	mA	1-3
F35	STOP	79	2.4	-11.7	-6.1	mA	1-3
F36	IR BIT 09	80	2.4	-0.6	0.02	mA	1-3
F37	IR BIT 08	82					
F38	IR BIT 07	83					
F39	CBDATA TMOUT	84	2.4	-0.6	0.02	mA	1-3
F40	BRKPT ABR 0	85	2.4	-0.2	0.02	mA	1-3
F41	SET ISTA EMP	87	2.4	-0.6	0.02	mA	1-3
F42	ISTACK FULL	89	2.4	-0.6	0.02	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F43	ISTA TO YREG	92	2.4	-1.2	0.04	mA	1-3
F44	ISTA TO PIPE	93	2.4	-1.2	0.04	mA	1-3
F45	RESET Y FULL	94	2.4	-0.6	0.02	mA	1-3
F46	J MAP OE	97	2.4	-2.0	0.05	mA	1-3
F47	IR BIT 41	98	2.4	-2.0	0.05	mA	1-3
F48	B/DESTN CLK	99	2.4	-0.6	0.02	mA	1-3

NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F48
Power supply current limit	1.73 A	NA	F1-F48
Forcing voltage, (Vih)	2.4 V	± 50 mV	F1-F48
Pattern voltage high, (VH)	3.0 V	± 100 mV	F1-F48
low, (VL)	0.0 V	± 100 mV	F1-F48

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA	± 31.0 uA	F1
	NA	± 62.0 uA	F2
	NA	± 31.0 uA	F3-F5
	NA	± 42.9 uA	F6
	NA	± 62.0 uA	F7,F8
	NA	± 307.0 uA	F9
	NA	± 360.0 uA	F10
	NA	± 304.0 uA	F11
	NA	± 31.0 uA	F12-F14
	NA	± 102.0 uA	F15
	NA	± 31.0 uA	F16,F17
	NA	± 62.0 uA	F18
	NA	± 13.7 uA	F19-F21
	NA	± 58.5 uA	F22
	NA	± 13.7 uA	F23
	NA	± 27.5 uA	F24
	NA	± 41.2 uA	F25
	NA	± 27.5 uA	F26
	NA	± 55.7 uA	F27
	NA	± 13.7 uA	F28
	NA	± 27.5 uA	F29
	NA	± 13.7 uA	F30-F34
	NA	± 280.0 uA	F35
	NA	± 31.0 uA	F36-F39
	NA	± 11.0 uA	F40
	NA	± 31.0 uA	F41,F42
	NA	± 62.0 uA	F43,F44
	NA	± 31.0 uA	F45
	NA	± 102.0 uA	F46,F47
	NA	± 31.0 uA	F48

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G1	IR BIT 11	3	0.4	-0.6	0.02	mA	1-3
G2	CB ADR TMOUT	4	0.4	-1.2	0.04	mA	1-3
G3	CLR2 RNK FUL	6	0.4	-0.6	0.02	mA	1-3
G4	CP MA CL SNC	7					
G5	CB ADDR TERM	8	0.4	-0.6	0.02	mA	1-3
G6	CB CYC ABORT	18	0.4	-1.4	-0.54	mA	1-3
G7	CB GRANT 3	21	0.4	-1.2	0.04	mA	1-3
G8	OTPT MDR FUL	26	0.4	-1.2	0.04	mA	1-3
G9	CB CLOCK 2	33	0.4	-6.0	0.15	mA	1-3
G10	CP CLOCK 1	39	0.4	-7.0	0.2	mA	1-3
G11	FL INSTR STA	41	0.4	-6.0	0.08	mA	1-3
G12	CB BUSY	44	0.4	-0.6	0.02	mA	1-3
G13	CB DATA TERM	45					
G14	RESET IN MDR	49	0.4	-0.6	0.02	mA	1-3
G15	NDRO SEL LAT	53	0.4	-2.0	0.05	mA	1-3
G16	WR LOCKOUT	54	0.4	-0.6	0.02	mA	1-3
G17	EXEC LOCKOUT	55	0.4	-0.6	0.02	mA	1-3
G18	READ LOCKOUT	56	0.4	-1.2	0.04	mA	1-3
G19	IR BIT 02	57	0.4	-0.25	0.025	mA	1-3
G20	IR BIT 01	58					
G21	IR BIT 00	59	0.4	-0.25	0.025	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G22	COM BUS CTRL	61	0.4	-1.1	0.07	mA	1-3
G23	IR BIT 10	62	0.4	-0.25	0.025	mA	1-3
G24	CB CTRL CD F	63	0.4	-0.5	0.05	mA	1-3
G25	FRC INS FTCH	64	0.4	-0.75	0.075	mA	1-3
G26	IR BIT 05	65	0.4	-0.5	0.05	mA	1-3
G27	KILL PIPE	66	0.4	-1.05	0.065	mA	1-3
G28	BUS BIT 00	67	0.4	-0.25	0.025	mA	1-3
G29	IR BIT 06	68	0.4	-0.5	0.05	mA	1-3
G30	IR BIT 04	69	0.4	-0.25	0.025	mA	1-3
G31	IR BIT 03	70					
G32	BYTE POINTER	71					
G33	EN PROTECT A	72					
G34	DEST CLK HLD	73	0.4	-0.25	0.025	mA	1-3
G35	STOP	79	0.4	-11.7	-6.1	mA	1-3
G36	IR BIT 09	80	0.4	-0.6	0.02	mA	1-3
G37	IR BIT 08	82					
G38	IR BIT 07	83					
G39	CBDATA TMOUT	84	0.4	-0.6	0.02	mA	1-3
G40	BRKPT ABR 0	85	0.4	-0.2	0.02	mA	1-3
G41	SET ISTA EMP	87	0.4	-0.6	0.02	mA	1-3
G42	ISTACK FULL	89	0.4	-0.6	0.02	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
G43	ISTA TO YREG	92	0.4	-1.2	0.04	mA	1-3
G44	ISTA TO PIPE	93	0.4	-1.2	0.04	mA	1-3
G45	RESET Y FULL	94	0.4	-0.6	0.02	mA	1-3
G46	J MAP OE	97	0.4	-2.0	0.05	mA	1-3
G47	IR BIT 41	98	0.4	-2.0	0.05	mA	1-3
G48	B/DESTN CLK	99	0.4	-0.6	0.02	mA	1-3

NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G48
Power supply current limit	1.73 A	NA	G1-G48
Forcing voltage, (Vil)	0.4 V	± 50 mV	G1-G48
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 G1-G48 G1-G48

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA	± 31.0 uA	G1
	NA	± 62.0 uA	G2
	NA	± 31.0 uA	G3-G5
	NA	± 43.0 uA	G6
	NA	± 62.0 uA	G7,G8
	NA	± 307.0 uA	G9
	NA	± 360.0 uA	G10
	NA	± 304.0 uA	G11
	NA	± 31.0 uA	G12-G14
	NA	± 102.0 uA	G15
	NA	± 31.0 uA	G16,G17
	NA	± 62.0 uA	G18
	NA	± 13.7 uA	G19-G21
	NA	± 58.5 uA	G22
	NA	± 13.7 uA	G23
	NA	± 27.5 uA	G24
	NA	± 41.2 uA	G25
	NA	± 27.5 uA	G26
	NA	± 55.7 uA	G27
	NA	± 13.7 uA	G28
	NA	± 27.5 uA	G29
	NA	± 13.7 uA	G30-G34
	NA	± 280.0 uA	G35
	NA	± 31.0 uA	G36-G39
	NA	± 11.0 uA	G40
	NA	± 31.0 uA	G41,G42
	NA	± 62.0 uA	G43,G44
	NA	± 31.0 uA	G45
	NA	± 102.0 uA	G46,G47
	NA	± 31.0 uA	G48

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H1	E START	2	-0.4	2.4	4.5	V	1-4
H2	ORIG RDY TP	5					
H3	EN PG W DATA	9					
H4	CB FUNC BIT0	11					
H5	CB FUNC BIT1	12					
H6	CB ORIG RDY	13					
H7	INSTR FETCH	14					
H8	CB FUNC BIT2	15					
H9	CB FUNC BIT3	16					
H10	CB WRITE EN	17					
H11	CB CYC ABORT	18					
H12	PAGING BUSY	19					
H13	EN CB AD/WR	20					
H14	(TP) 1	22					
H15	AD BUS MASTR	23					
H16	CB REQ3 (CP)	27					
H17	2ND RNK FULL	28					
H18	(TP) 2	29					
H19	CLK 3RD RANK	30					
H20	CLK IN MDR	32					
H21	(TP) 3	34	-0.4	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
H22	CB CLK BUF	35	-0.4	2.4	4.5	V	1-4
H23	(TP) 4	36					
H24	TNT DT BS MR	37					
H25	DT BS MASTR	38					
H26	CLK Y	42					
H27	LD PG B15-12	43					
H28	EN INCR PC	46					
H29	CLK BRKPT ST	47					
H30	CLK PIPE	48					
H31	IN MDR FULL	50					
H32	KIL PI 4 RNK	52					
H33	SEL ISTA/SEL	74					
H34	RESUME ERROR	76					
H35	BRKPT 4 RANK	77	-0.4	2.4	4.5	V	1-4
H36	PTCT ER 4 RK	78	-0.16	2.4	4.5	V	1-4
H37	CLOCK ISTACK	86	-0.4	2.4	4.5	V	1-4
H38	INCR ISTA WR	88	-0.4	2.4	4.5	V	1-4
H39	2 WD READ	91	-0.16	2.4	4.5	V	1-4
H40	Y REG FULL	95	-0.4	2.4	4.5	V	1-4
H41	PIPE FULL	96					
H42	INCR ISTA RD	100	-0.4	2.4	4.5	V	1-4

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NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	H1-H42
Power supply current limit	1.73 A	NA	H1-H42
Forcing current, (Ioh)	-0.4 mA -0.16 mA -0.4 mA -0.16 mA -0.4 mA	± 40 uA ± 16 uA ± 40 uA ± 16 uA ± 40 uA	H1-H35 H36 H37,H38 H39 H40-H42
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 H1-H42 H1-H42
Output voltage measurement	NA	± 105 mV	H1-H42

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

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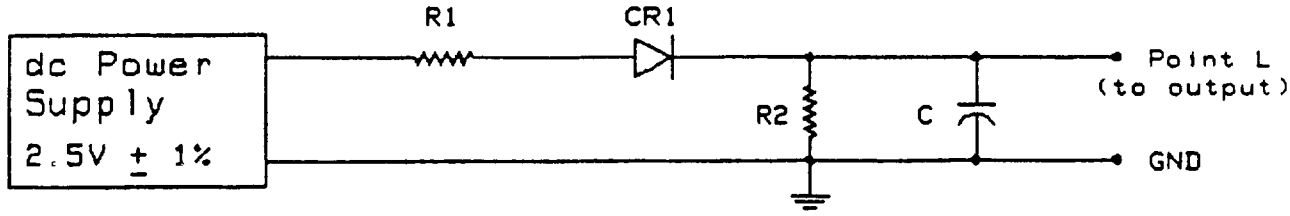
NOTES: - Continued.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table IX and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 100 ± 10 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(c) The functional test shall be performed with only a 100 ± 10 pF load (including probe and parasitic capacitance) attached to each output.

(d) C not required for any other tests.

FIGURE 9. Logic load circuit.

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NOTES: - Continued.

3. Select R1 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are in the table below.

4. The value of R2 for each output load shall be as specified in the table below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
2,5,9,11,12,13, 14,15,16,17,18, 19,20,22,23,27, 28,29,30,32,34, 35,36,37,38,42, 43,46,47,48,50, 52,74,76,77,86, 88,95,96,100	6200	6.0	0.5
78,91	16000	1.6	0.5

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE X. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J1	E START	2	6.0	0.0	0.5	V	1-4
J2	ORIG RDY TP	5					
J3	EN PG W DATA	9					
J4	CB FUNC BIT0	11					
J5	CB FUNC BIT1	12					
J6	CB ORIG RDY	13					
J7	INSTR FETCH	14					
J8	CB FUNC BIT2	15					
J9	CB FUNC BIT3	16					
J10	CB WRITE EN	17					
J11	CB CYC ABORT	18					
J12	PAGING BUSY	19					
J13	EN CB AD/WR	20					
J14	(TP) 1	22					
J15	AD BUS MASTR	23					
J16	CB REQ3 (CP)	27					
J17	2ND RNK FULL	28					
J18	(TP) 2	29					
J19	CLK 3RD RANK	30					
J20	CLK IN MDR	32					
J21	(TP) 3	34	6.0	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J22	CB CLK BUF	35	6.0	0.0	0.5	V	1-4
J23	(TP) 4	36					
J24	TNT DT BS MR	37					
J25	DT BS MASTR	38					
J26	CLK Y	42					
J27	LD PG B15-12	43					
J28	EN INCR PC	46					
J29	CLK BRKPT ST	47					
J30	CLK PIPE	48					
J31	IN MDR FULL	50					
J32	KIL PI 4 RNK	52					
J33	SEL ISTA/SEL	74					
J34	RESUME ERROR	76					
J35	BRKPT 4 RANK	77	6.0	0.0	0.5	V	1-4
J36	PTCT ER 4 RK	78	1.6	0.0	0.5	V	1-4
J37	CLOCK ISTACK	86	6.0	0.0	0.5	V	1-4
J38	INCR ISTA WR	88	6.0	0.0	0.5	V	1-4
J39	2 WD READ	91	1.6	0.0	0.5	V	1-4
J40	Y REG FULL	95	6.0	0.0	0.5	V	1-4
J41	PIPE FULL	96					
J42	INCR ISTA RD	100	6.0	0.0	0.5	V	1-4

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NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J42
Power supply current limit	1.73 A	NA	J1-J42
Forcing current, (Iol)	6.0 mA 1.6 mA 6.0 mA 1.6 mA 6.0 mA	± 600 uA ± 160 uA ± 600 uA ± 160 uA ± 600 uA	J1-J35 J36 J37,J38 J39 J40-J42
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 J1-J42 J1-J42
Output voltage measurement	NA	± 25 mV	J1-J42

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

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NOTES: - Continued.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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TABLE XI. Output leakage current, high and low level, test.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
High level							
K1	CB FUNC BIT0	11	2.4	-0.05	0.05	mA	1-3
K2	CB FUNC BIT1	12					
K3	CB ORIG RDY	13					
K4	INSTR FETCH	14					
K5	CB FUNC BIT2	15					
K6	CB FUNC BIT3	16					
K7	CB WRITE EN	17	2.4	-0.05	0.05	mA	1-3
Low level							
K8	CB FUNC BIT0	11	0.4	-0.05	0.05	mA	1-3
K9	CB FUNC BIT1	12					
K10	CB ORIG RDY	13					
K11	INSTR FETCH	14					
K12	CB FUNC BIT2	15					
K13	CB FUNC BIT3	16					
K14	CB WRITE EN	17	0.4	-0.05	0.05	mA	1-3

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NOTES:

1. Output leakage current, high and low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	K1-K14
Power supply current limit	1.73 A	NA	K1-K14
Forcing voltage, (Voh) (Vol)	 2.4 V 0.4 V	 ± 50 mV ± 50 mV	 K1-K7 K8-K14
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 K1-K14 K1-K14
Output current measurement	NA	± 5 uA	K1-K14

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

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NOTES: - Continued.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XI and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XII. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L1	IR BIT 11	3	-18.0	-1.2	0.0	V	1-3
L2	CB ADR TMOUT	4					
L3	CLR2 RNK FUL	6					
L4	CP MA CL SNC	7					
L5	CB ADDR TERM	8					
L6	CB GRANT 3	21					
L7	OTPT MDR FUL	26					
L8	CB CLOCK 2	33					
L9	CP CLOCK 1	39					
L10	FL INSTR STA	41					
L11	CB BUSY	44					
L12	CB DATA TERM	45					
L13	RESET IN MDR	49					
L14	NDRO SEL LAT	53					
L15	WR LOCKOUT	54					
L16	EXEC LOCKOUT	55					
L17	READ LOCKOUT	56	-18.0	-1.2	0.0	V	1-3
L18	IR BIT 02	57	-18.0	-1.5	0.0	V	1-3
L19	IR BIT 01	58					
L20	IR BIT 00	59	-18.0	-1.5	0.0	V	1-3
L21	COM BUS CTRL	61	-18.0	-1.2	0.0	V	1-3

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TABLE XII. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L22	IR BIT 10	62	-18.0	-1.5	0.0	V	1-3
L23	CB CTRL CD F	63					
L24	FRC INS FTCH	64					
L25	IR BIT 05	65	-18.0	-1.5	0.0	V	1-3
L26	KILL PIPE	66	-18.0	-1.2	0.0	V	1-3
L27	BUS BIT 00	67	-18.0	-1.5	0.0	V	1-3
L28	IR BIT 06	68					
L29	IR BIT 04	69					
L30	IR BIT 03	70					
L31	BYTE POINTER	71					
L32	EN PROTECT A	72					
L33	DEST CLK HLD	73	-18.0	-1.5	0.0	V	1-3
L34	IR BIT 09	80	-18.0	-1.2	0.0	V	1-3
L35	IR BIT 08	82					
L36	IR BIT 07	83					
L37	CBDATA TMOUT	84	-18.0	-1.2	0.0	V	1-3
L38	BRKPT ABR 0	85	-18.0	-1.5	0.0	V	1-3
L39	SET ISTA EMP	87	-18.0	-1.2	0.0	V	1-3
L40	ISTACK FULL	89					
L41	ISTA TO YREG	92					
L42	ISTA TO PIPE	93	-18.0	-1.2	0.0	V	1-3

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TABLE XII. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L43	RESET Y FULL	94	-18.0	-1.2	0.0	V	1-3
L44	J MAP OE	97					
L45	IR BIT 41	98					
L46	B/DESTN CLK	99	-18.0	-1.2	0.0	V	1-3

NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	Off	NA	L1-L46
Power supply current limit	NA	NA	L1-L46
Forcing current, (Iin)	-18 mA	± 1.8 mA	L1-L46
Pattern voltage high, (VH) low, (VL)	 NA NA	 NA NA	 L1-L46 L1-L46

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input voltage measurement	NA	± 60 mV	L1-L17
	NA	± 75 mV	L18-L20
	NA	± 60 mV	L21
	NA	± 75 mV	L22-L25
	NA	± 60 mV	L26
	NA	± 75 mV	L27-L33
	NA	± 60 mV	L34-L37
	NA	± 75 mV	L38
	NA	± 60 mV	L39-L46

2. The following procedure is required to perform this test:

(a) No power shall be applied to the module.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XII and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE XIII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
M1	PSFC	1	3.7	6.9	uF	1-4
M2	PSFC	51	3.1	5.8	uF	1-4

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	M1,M2
Capacitance bridge voltage	0.3 Vp	± 30 mVp	M1,M2
Capacitance measurement	NA NA	± 160 nF ± 135 nF	M1 M2

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XIII.

3. The test circuit shall be as indicated on figure 10.

4. Abbreviations used are defined in 6.1.

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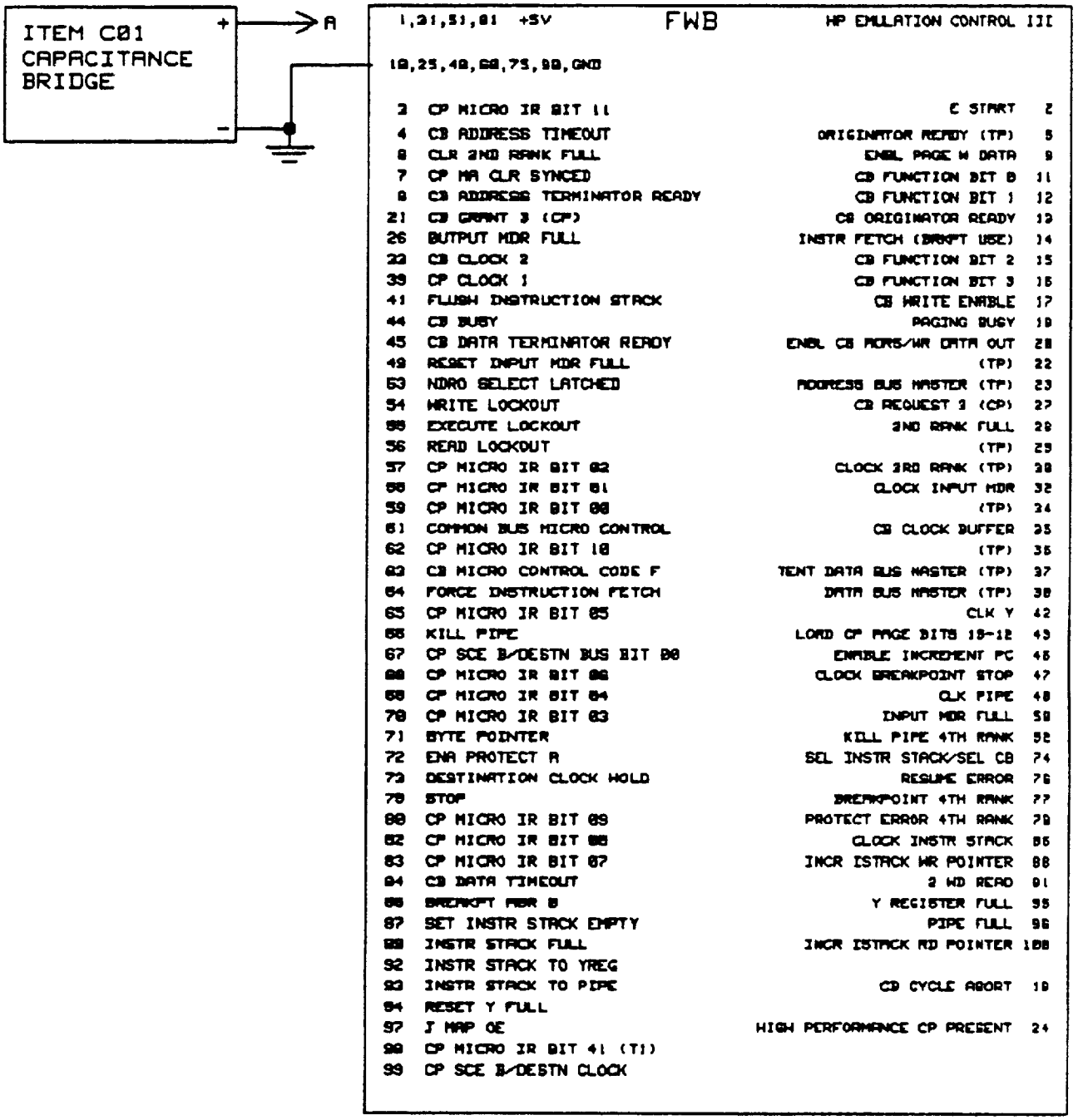


FIGURE 10. Power supply filter capacitance test circuit for table XIII.

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TABLE XIV. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N	Icc	1,31, 51,81	480	1150	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N
Power supply current limit	1.73 A	NA	N
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 N N
Power supply current measurement	NA	± 32.5 mA	N

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIV.

3. Abbreviations used are defined in 6.1.

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TABLE XV. Functional test.

Test cond	Test description	Notes	Figs
P1	Functional test minimum power supply voltage maximum pattern rate (225 nS)	1-3	9,11
P2	Functional test maximum power supply voltage maximum pattern rate (225 nS)	1-3	9,11

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	P1 P2
Power supply current limit	1.73 A	NA	P1,P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105 mV ± 155 mV ± 25 mV	 P1 P2 P1,P2

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Check the source to verify that this is the current version before use.

NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load of 100 pF (see figure 9). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.5\ V$

(c) The module inputs shall not violate the minimum input timing parameters referenced on figure 11.

3. Abbreviations used are defined in 6.1.

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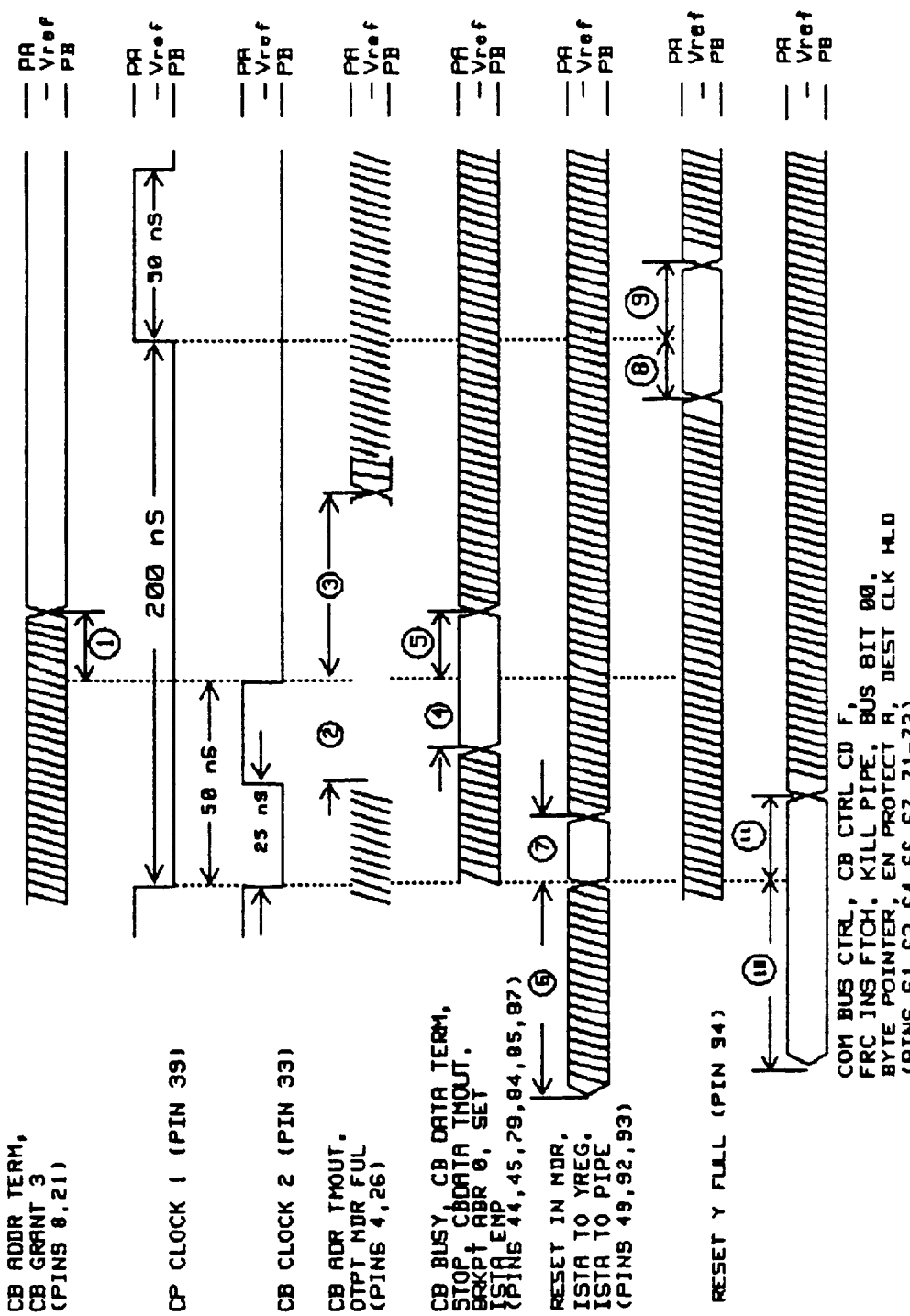


FIGURE 11. Input timing parameters for table XV.

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NOTES:

1. (1) CB CLOCK 2 to CB ADDR TERM, CB GRANT 3. Hold time: minimum 50 nS.

2. (2) CB ADR TMOUT, OTPT MDR FUL to CB CLOCK 2. Setup time: minimum 20 nS.

3. (3) CB CLOCK 2 to CB ADR TMOUT, OTPT MDR FUL. Hold time: minimum 60 nS.

4. (4) CB BUSY, CB DATA TERM, STOP, CBDATA TMOUT, BRKPT ABR 0, SET ISTA EMP to CB CLOCK 2. Setup time: minimum 10 nS.

5. (5) CB CLOCK 2 to CB BUSY, CB DATA TERM, STOP, CBDATA TMOUT, BRKPT ABR 0, SET ISTA EMP. Hold time: minimum 50 nS.

6. (6) RESET IN MDR, ISTA TO YREG, ISTA TO PIPE to CP CLOCK 1. Setup time: minimum 100 nS.

7. (7) CP CLOCK 1 to RESET IN MDR, ISTA TO YREG, ISTA TO PIPE. Hold time: minimum 20 nS.

8. (8) RESET Y FULL to CP CLOCK 1. Setup time: minimum 25 nS.

9. (9) CP CLOCK 1 to RESET Y FULL. Hold time: minimum 15 nS.

10. (10) COM BUS CTRL, CB CTRL CD F, FRC INS FTCH, KILL PIPE, BUS BIT 00, BYTE POINTER, EN PROTECT A, DEST CLK HLD to CP CLOCK 1. Setup time: minimum 100 nS.

11. (11) CP CLOCK 1 to COM BUS CTRL, CB CTRL CD F, FRC INS FTCH, KILL PIPE, BUS BIT 00, BYTE POINTER, EN PROTECT A, DEST CLK HLD. Hold time: minimum 20 nS.

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TABLE XVI. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate (225 nS) minimum power supply	1-6
Q2	Dynamic test, sample maximum pattern rate (225 nS) maximum power supply	1-6
Q3	Dynamic test, qualification maximum pattern rate (225 nS) minimum power supply	1-6
Q4	Dynamic test, qualification maximum pattern rate (225 nS) maximum power supply	1-6

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Current limit	Test condition
Power supply (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	1.73 A 1.73 A	Q1,Q3 Q2,Q4
Input voltage high, (VH) low, (VL)	2.4 V 0.4 V	± 100 mV ± 100 mV	NA NA	Q1-Q4 Q1-Q4

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NOTES: - Continued.

Condition	Value	Accuracy	Current limit	Test condition
Output voltage verification				
high, (Voh)	NA	± 100 mV	NA	Q1,Q3
high, (Voh)	NA	± 100 mV	NA	Q2,Q4
low, (Vol)	NA	± 100 mV	NA	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

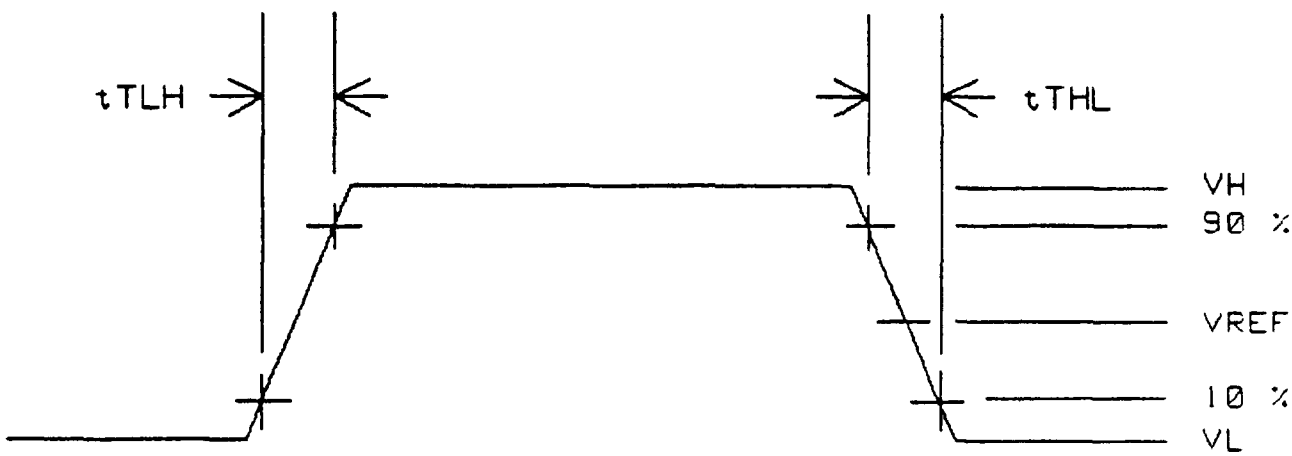
(b) The output compare levels shall be as follows:

High level: Vref ≤ Voh ≤ Vcc
Low level: 0.0 V ≤ Vol ≤ Vref

(c) The dynamic parameters are found in table XVII.

3. The load circuit for this test is specified on figure 9.

4. Input/Output (I/O) waveforms are as referenced on figure 12. All of the input waveforms shall have the following characteristics:



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NOTES: - Continued.

Where:

(a) VH and VL are as specified in the test tolerance table.

(b) tTLH = tTHL $\leq$ 10 nS.

(c) Vref = 1.6 V.

5. This test shall be performed as indicated in table XVI.

6. Abbreviations used are defined in 6.1.

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TABLE XVII. Dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	CB CLOCK 2 to EN CB AD/WR	-	-	-	70	70	70	nS	12
1	tpd	CB CLOCK 2 to RESUME ERROR	-	-	-	70	70	70	nS	12
1	tpd	CB CLOCK 2 to PIPE FULL	-	-	-	70	70	70	nS	12
2	tpd	CB GRANT 3 to CB FUNC BIT0	-	-	-	50	50	50	nS	12
2	tpd	CB GRANT 3 to CB FUNC BIT1	-	-	-	50	50	50	nS	12
2	tpd	CB GRANT 3 to CB FUNC BIT2	-	-	-	50	50	50	nS	12
2	tpd	CB GRANT 3 to CB FUNC BIT3	-	-	-	50	50	50	nS	12
2	tpd	CB GRANT 3 to INSTR FETCH	-	-	-	50	50	50	nS	12
2	tpd	CB GRANT 3 to CB CYC ABORT	-	-	-	50	50	50	nS	12
3	tpd	B/DESTN CLK to PAGING BUSY	-	-	-	60	60	60	nS	12
4	tpd	CB CLOCK 2 to CB REQ3 (CP)	-	-	-	60	60	60	nS	12
4	tpd	CB CLOCK 2 to 2ND RNK FULL	-	-	-	60	60	60	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
4	tpd	CB CLOCK 2 to INCR ISTA WR	-	-	-	60	60	60	nS	12
4	tpd	CB CLOCK 2 to Y REG FULL	-	-	-	60	60	60	nS	12
5	tpd	CB CLOCK 2 to EN PG W DATA	-	-	-	40	40	40	nS	12
6	tpd	CB CLOCK 2 to CLK IN MDR	-	-	-	50	50	50	nS	12
6	tpd	CB CLOCK 2 to CLK Y	-	-	-	50	50	50	nS	12
6	tpd	CB CLOCK 2 to LD PG B15-12	-	-	-	50	50	50	nS	12
6	tpd	CB CLOCK 2 to CLK BRKPT ST	-	-	-	50	50	50	nS	12
6	tpd	CB CLOCK 2 to CLK PIPE	-	-	-	50	50	50	nS	12
7	tpd	CP CLOCK 1 to IN MDR FULL	-	-	-	100	100	100	nS	12
7	tpd	CP CLOCK 1 to SEL ISTA/SEL	-	-	-	100	100	100	nS	12
7	tpd	CP CLOCK 1 to 2 WD READ	-	-	-	100	100	100	nS	12
7	tpd	CP CLOCK 1 to INCR ISTA RD	-	-	-	100	100	100	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
8	tpd	CB CLOCK 2 to KIL PI 4 RNK	-	-	-	110	110	110	nS	12
8	tpd	CB CLOCK 2 to PTCT ER 4 RK	-	-	-	110	110	110	nS	12
8	tpd	CB CLOCK 2 to BRKPT 4 RANK	-	-	-	110	110	110	nS	12
9	tpd	ISTACK FULL to CLOCK ISTACK	-	-	-	20	20	20	nS	12
10	th	CB CLOCK 2 to CB ADDR TERM	50	50	50	-	-	-	nS	12
10	th	CB CLOCK 2 to CB GRANT 3	50	50	50	-	-	-	nS	12
11	tsu	CB ADR TMOUT to CB CLOCK 2	20	20	20	-	-	-	nS	12
11	tsu	OTPT MDR FUL to CB CLOCK 2	20	20	20	-	-	-	nS	12
12	th	CB CLOCK 2 to CB ABR TMOUT	60	60	60	-	-	-	nS	12
12	th	CB CLOCK 2 to OTPT MDR FUL	60	60	60	-	-	-	nS	12
13	tsu	CB BUSY to CB CLOCK 2	10	10	10	-	-	-	nS	12
13	tsu	CB DATA TERM to CB CLOCK 2	10	10	10	-	-	-	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
13	tsu	STOP to CB CLOCK 2	10	10	10	-	-	-	nS	12
13	tsu	CBDATA TMOUT to CB CLOCK 2	10	10	10	-	-	-	nS	12
13	tsu	BRKPT ABR 0 to CB CLOCK 2	10	10	10	-	-	-	nS	12
13	tsu	SET ISTA EMP to CB CLOCK 2	10	10	10	-	-	-	nS	12
14	th	CB CLOCK 2 to CB BUSY	50	50	50	-	-	-	nS	12
14	th	CB CLOCK 2 to CB DATA TERM	50	50	50	-	-	-	nS	12
14	th	CB CLOCK 2 to STOP	50	50	50	-	-	-	nS	12
14	th	CB CLOCK 2 to CBDATA TMOUT	50	50	50	-	-	-	nS	12
14	th	CB CLOCK 2 to BRKPT ABR 0	50	50	50	-	-	-	nS	12
14	th	CB CLOCK 2 to SET ISTA EMP	50	50	50	-	-	-	nS	12
15	tsu	RESET IN MDR to CP CLOCK 1	100	100	100	-	-	-	nS	12
15	tsu	ISTA TO YREG to CP CLOCK 1	100	100	100	-	-	-	nS	12
15	tsu	ISTA TO PIPE to CP CLOCK 1	100	100	100	-	-	-	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
16	th	CP CLOCK 1 to RESET IN MDR	20	20	20	-	-	-	nS	12
16	th	CP CLOCK 1 to ISTA TO YREG	20	20	20	-	-	-	nS	12
16	th	CP CLOCK 1 to ISTA TO PIPE	20	20	20	-	-	-	nS	12
17	tsu	RESET Y FULL to CP CLOCK 1	25	25	25	-	-	-	nS	12
18	th	CP CLOCK 1 to RESET Y FULL	15	15	15	-	-	-	nS	12
19	tsu	COM BUS CTRL to CP CLOCK 1	100	100	100	-	-	-	nS	12
19	tsu	CB CTRL CD F to CP CLOCK 1	100	100	100	-	-	-	nS	12
19	tsu	FRC INS FTCH to CP CLOCK 1	100	100	100	-	-	-	nS	12
19	tsu	KILL PIPE to CP CLOCK 1	100	100	100	-	-	-	nS	12
19	tsu	BUS BIT 00 CP CLOCK 1	100	100	100	-	-	-	nS	12
19	tsu	BYTE POINTER to CP CLOCK 1	100	100	100	-	-	-	nS	12
19	tsu	EN PROTECT A to CP CLOCK 1	100	100	100	-	-	-	nS	12

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Fig
			-55°C	25°C	85°C	-55°C	25°C	85°C		
19	tsu	DEST CLK HLD to CP CLOCK 1	100	100	100	-	-	-	nS	12
20	th	CP CLOCK 1 to COM BUS CTRL	20	20	20	-	-	-	nS	12
20	th	CP CLOCK 1 to CB CTRL CD F	20	20	20	-	-	-	nS	12
20	th	CP CLOCK 1 to FRC INS FTCH	20	20	20	-	-	-	nS	12
20	th	CP CLOCK 1 to KILL PIPE	20	20	20	-	-	-	nS	12
20	th	CP CLOCK 1 to BUS BIT 00	20	20	20	-	-	-	nS	12
20	th	CP CLOCK 1 to BYTE POINTER	20	20	20	-	-	-	nS	12
20	th	CP CLOCK 1 to EN PROTECT A	20	20	20	-	-	-	nS	12
20	th	CP CLOCK 1 to DEST CLK HLD	20	20	20	-	-	-	nS	12

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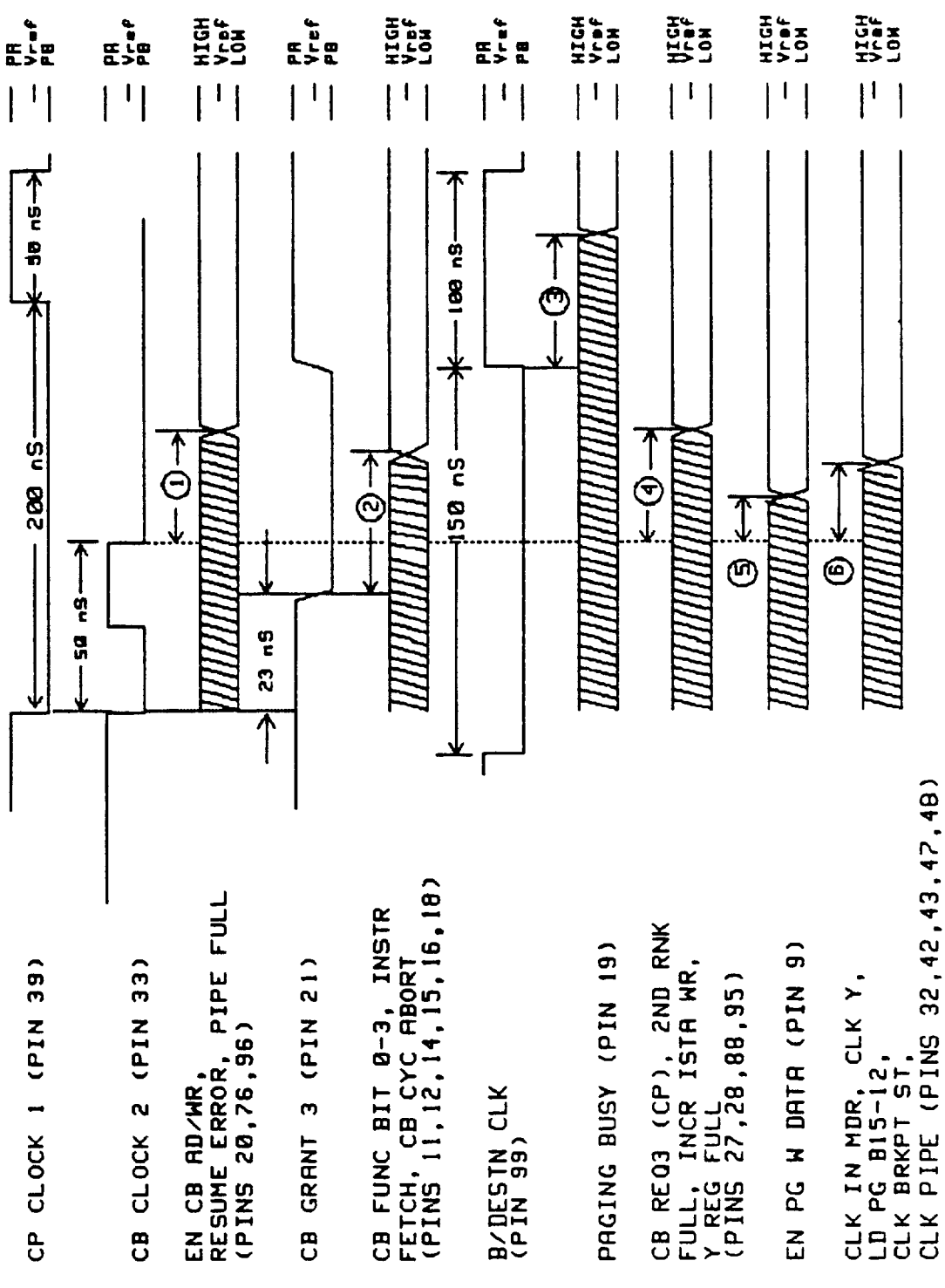


FIGURE 12a. Dynamic waveforms for table XVII.

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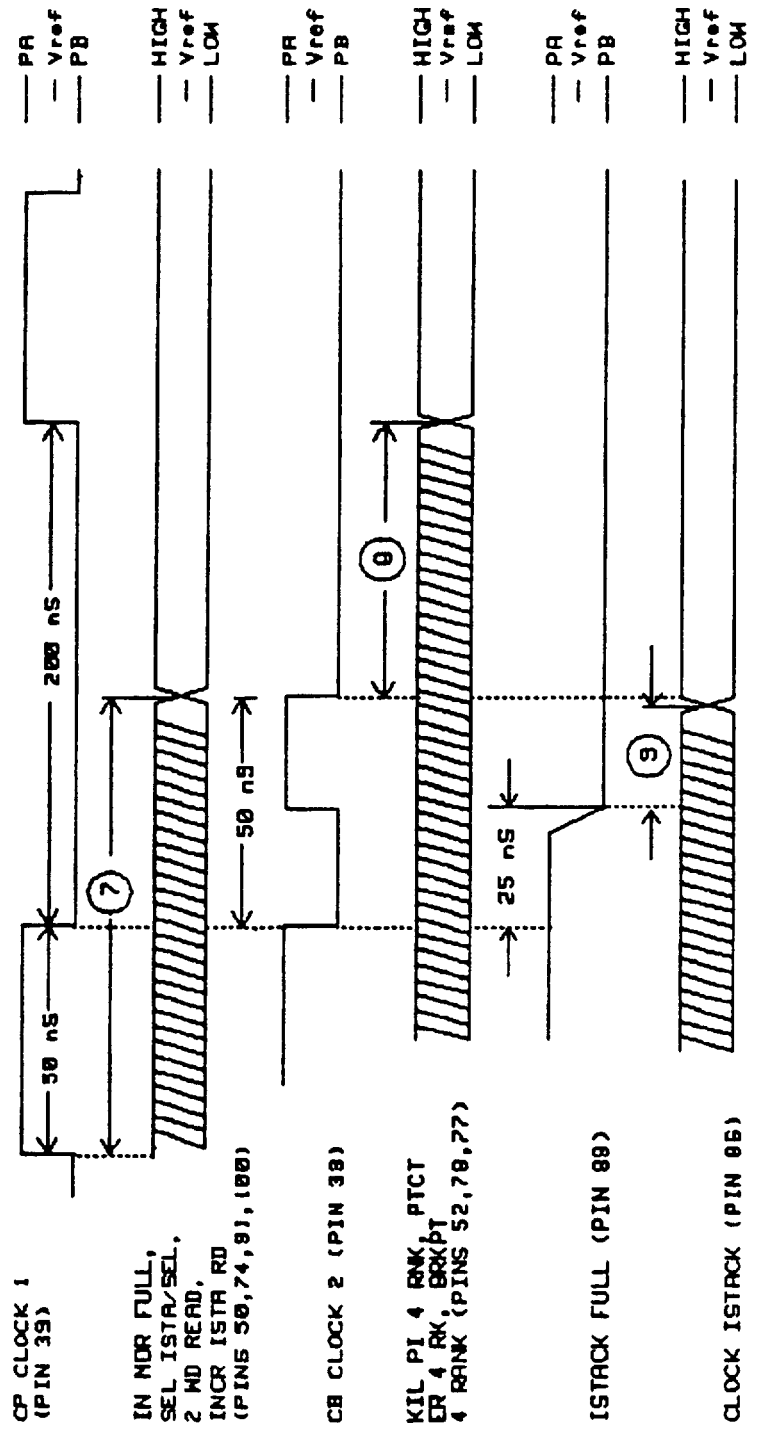


FIGURE 12b. Dynamic waveforms for table XVII.

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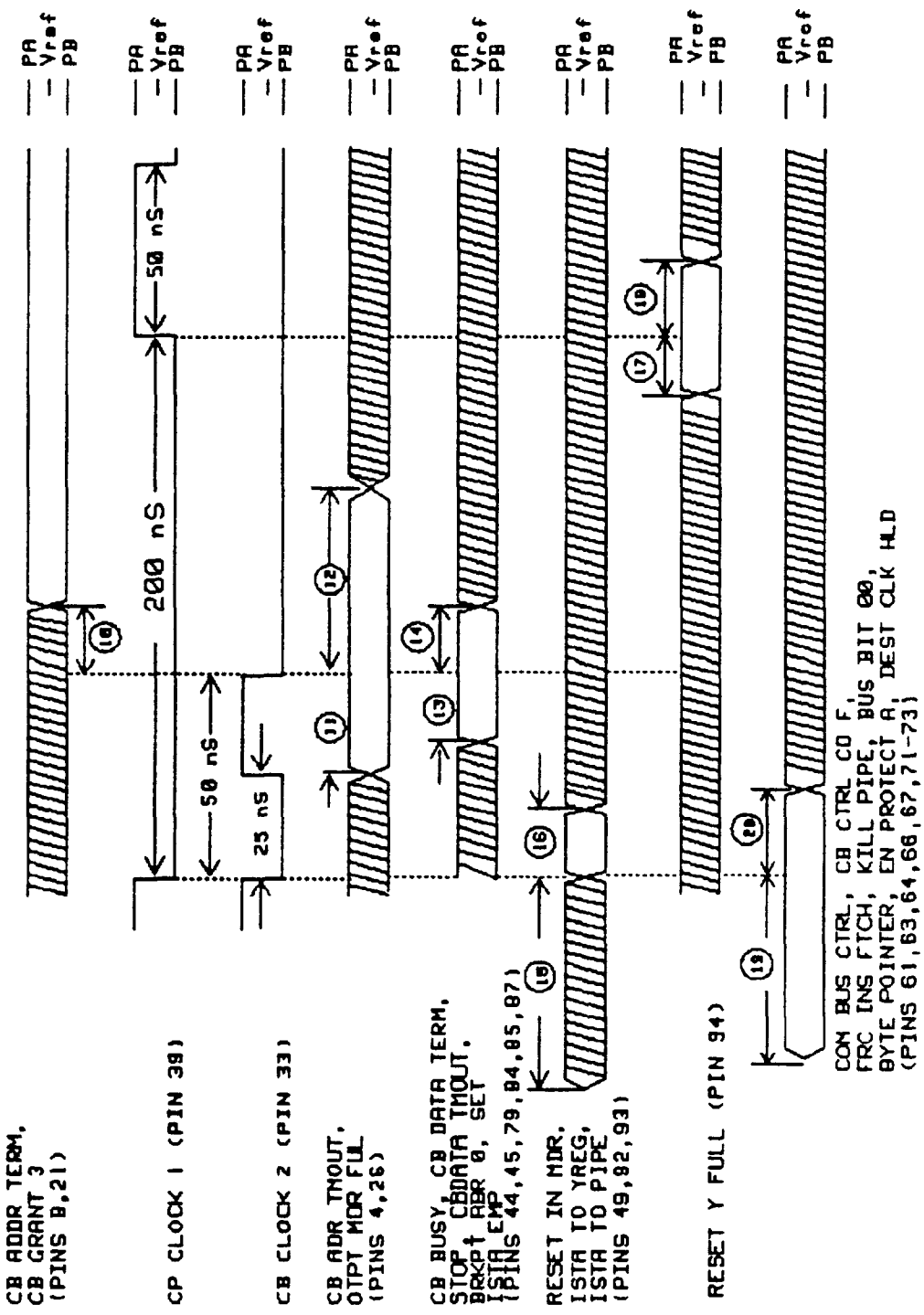


FIGURE 12c. Dynamic waveforms for table XVII.

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TABLE XVIII. Input and output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
Inputs						
R1	IR BIT 11	3	2	75	pF	1-4
R2	CB ADR TMOUT	4	4	150	pF	1-4
R3	CLR2 RNK FUL	6	2	75	pF	1-4
R4	CP MA CL SNC	7	2	50	pF	1-4
R5	CB ADDR TERM	8	2	75	pF	1-4
R6	CB GRANT 3	21	4	75	pF	1-4
R7	OTPT MDR FUL	26	4	75	pF	1-4
R8	CB CLOCK 2	33	6	75	pF	1-4
R9	CP CLOCK 1	39	10	175	pF	1-4
R10	FL INSTR STA	41	4	100	pF	1-4
R11	CB BUSY	44	2	50	pF	1-4
R12	CB DATA TERM	45	2	75	pF	1-4
R13	RESET IN MDR	49	2	75	pF	1-4
R14	NDRO SEL LAT	53	2	50	pF	1-4
R15	WR LOCKOUT	54	2	50	pF	1-4
R16	EXEC LOCKOUT	55	2	75	pF	1-4
R17	READ LOCKOUT	56	4	75	pF	1-4
R18	IR BIT 02	57	2	50	pF	1-4
R19	IR BIT 00	59	2	50	pF	1-4
R20	COM BUS CTRL	61	6	125	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R21	IR BIT 10	62	2	75	pF	1-4
R22	CB CTRL CD F	63	4	75	pF	1-4
R23	FRC INS FTCH	64	6	100	pF	1-4
R24	IR BIT 05	65	4	75	pF	1-4
R25	KILL PIPE	66	6	125	pF	1-4
R26	BUS BIT 00	67	2	75	pF	1-4
R27	IR BIT 06	68	4	100	pF	1-4
R28	IR BIT 04	69	2	50	pF	1-4
R29	IR BIT 03	70				
R30	BYTE POINTER	71	2	50	pF	1-4
R31	EN PROTECT A	72	2	75	pF	1-4
R32	DEST CLK HLD	73	2	100	pF	1-4
R33	STOP	79	4	75	pF	1-4
R34	IR BIT 09	80	2	50	pF	1-4
R35	IR BIT 08	82				
R36	IR BIT 07	83	2	50	pF	1-4
R37	CBDATA TMOUT	84	2	75	pF	1-4
R38	BRKPT ABR 0	85	2	75	pF	1-4
R39	SET ISTA EMP	87	2	50	pF	1-4
R40	ISTACK FULL	89	2	50	pF	1-4
R41	ISTA TO YREG	92	4	75	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R42	ISTA TO PIPE	93	4	100	pF	1-4
R43	RESET Y FULL	94	2	50	pF	1-4
R44	J MAP OE	97	2	125	pF	1-4
R45	IR BIT 41	98	2	125	pF	1-4
R46	B/DESTN CLK	99	2	75	pF	1-4
Outputs						
R47	ORIG RDY TP	5	6	125	pF	1-4
R48	CB FUNC BIT0	11	2	75	pF	1-4
R49	CB FUNC BIT1	12				
R50	CB ORIG RDY	13				
R51	INSTR FETCH	14				
R52	CB FUNC BIT2	15				
R53	CB FUNC BIT3	16				
R54	CB WRITE EN	17	2	75	pF	1-4
R55	EN CB AD/WR	20	6	100	pF	1-4
R56	CB CLK BUF	35	2	50	pF	1-4
R57	EN INCR PC	46	2	100	pF	1-4
Bidirectional						
R58	CB CYC ABORT	18	6	100	pF	1-4
Input						
R59	IR BIT 01	58	2	50	pF	1-4

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NOTES:

1. Input and output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	R1-R59
Capacitance meter voltage	15 mV	± 1.5 mV	R1-R59
Capacitance measurement	NA	± 3.65 pF	R1
	NA	± 7.3 pF	R2
	NA	± 3.65 pF	R3
	NA	± 2.4 pF	R4
	NA	± 3.65 pF	R5
	NA	± 3.55 pF	R6,R7
	NA	± 3.45 pF	R8
	NA	± 8.25 pF	R9
	NA	± 4.8 pF	R10
	NA	± 2.4 pF	R11
	NA	± 3.65 pF	R12,R13
	NA	± 2.4 pF	R14,R15
	NA	± 3.65 pF	R16
	NA	± 3.55 pF	R17
	NA	± 2.4 pF	R18,R19
	NA	± 5.95 pF	R20
	NA	± 3.65 pF	R21
	NA	± 3.55 pF	R22
	NA	± 4.7 pF	R23
	NA	± 3.55 pF	R24
	NA	± 5.95 pF	R25
	NA	± 3.65 pF	R26
	NA	± 4.8 pF	R27
	NA	± 2.4 pF	R28-R30

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Capacitance measurement	NA	± 3.65 pF	R31
	NA	± 4.9 pF	R32
	NA	± 3.55 pF	R33
	NA	± 2.4 pF	R34-R36
	NA	± 3.65 pF	R37,R38
	NA	± 2.4 pF	R39,R40
	NA	± 3.55 pF	R41
	NA	± 4.8 pF	R42
	NA	± 2.4 pF	R43
	NA	± 6.15 pF	R44,R45
	NA	± 3.65 pF	R46
	NA	± 5.95 pF	R47
	NA	± 3.65 pF	R48-R54
	NA	± 4.7 pF	R55
	NA	± 2.4 pF	R56
	NA	± 4.9 pF	R57
	NA	± 4.7 pF	R58
	NA	± 2.4 pF	R59

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

3. The test circuit shall be as indicated on figure 13.

4. Abbreviations used are defined in 6.1.

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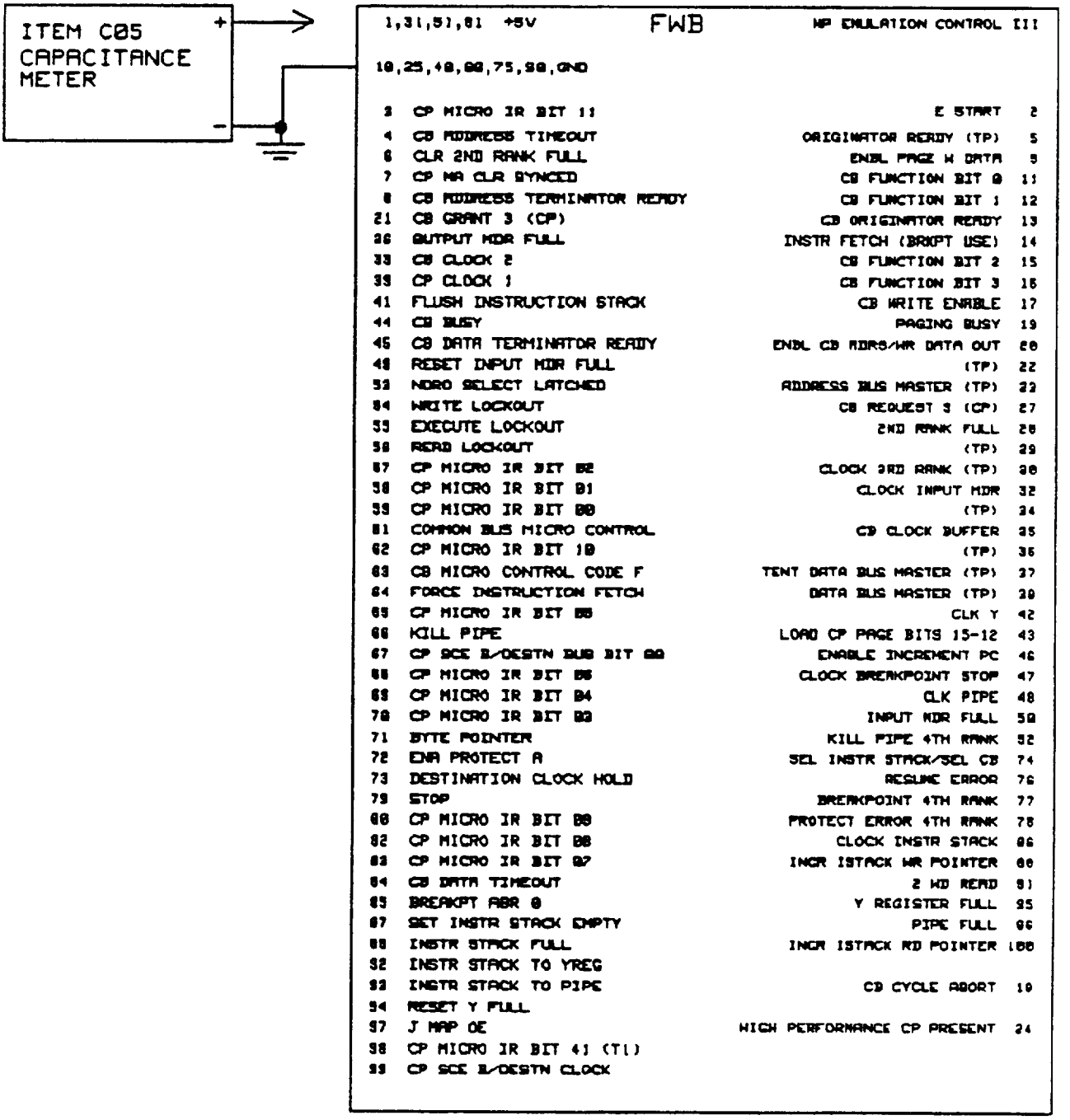


FIGURE 13. Input and output capacitance test circuit for table XVIII.

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TABLE XIX. Power supply transient amplitude test.

Test cond	Parameter	Pin number	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
T1	PSTA	1	0	400	mV	1-4	14,15
T2	PSTA	51	0	400	mV	1-4	14,15

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	T1,T2
Power supply current limit	1.73 A	NA	T1,T2
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 T1,T2 T1,T2
Power supply transient amplitude measurement	NA	± 20 mV	T1,T2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XIX.

(c) Vcc pins 1 and 31 shall be shorted together, pins 51 and 81 shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XIX.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 14 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. The test circuit shall be as indicated on figure 15.

4. Abbreviations used are defined in 6.1.

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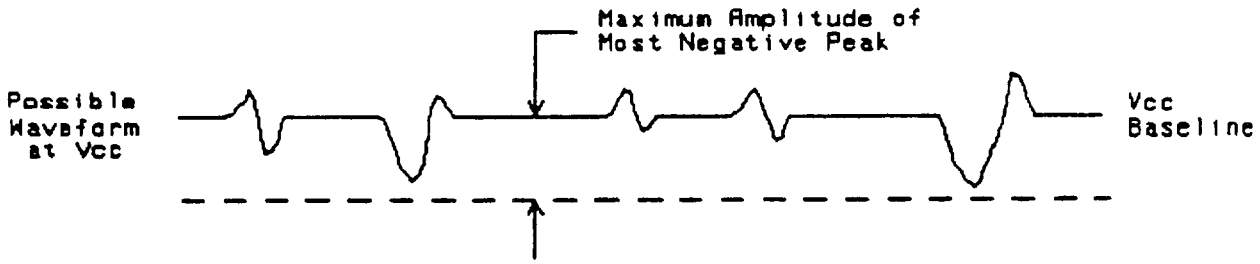
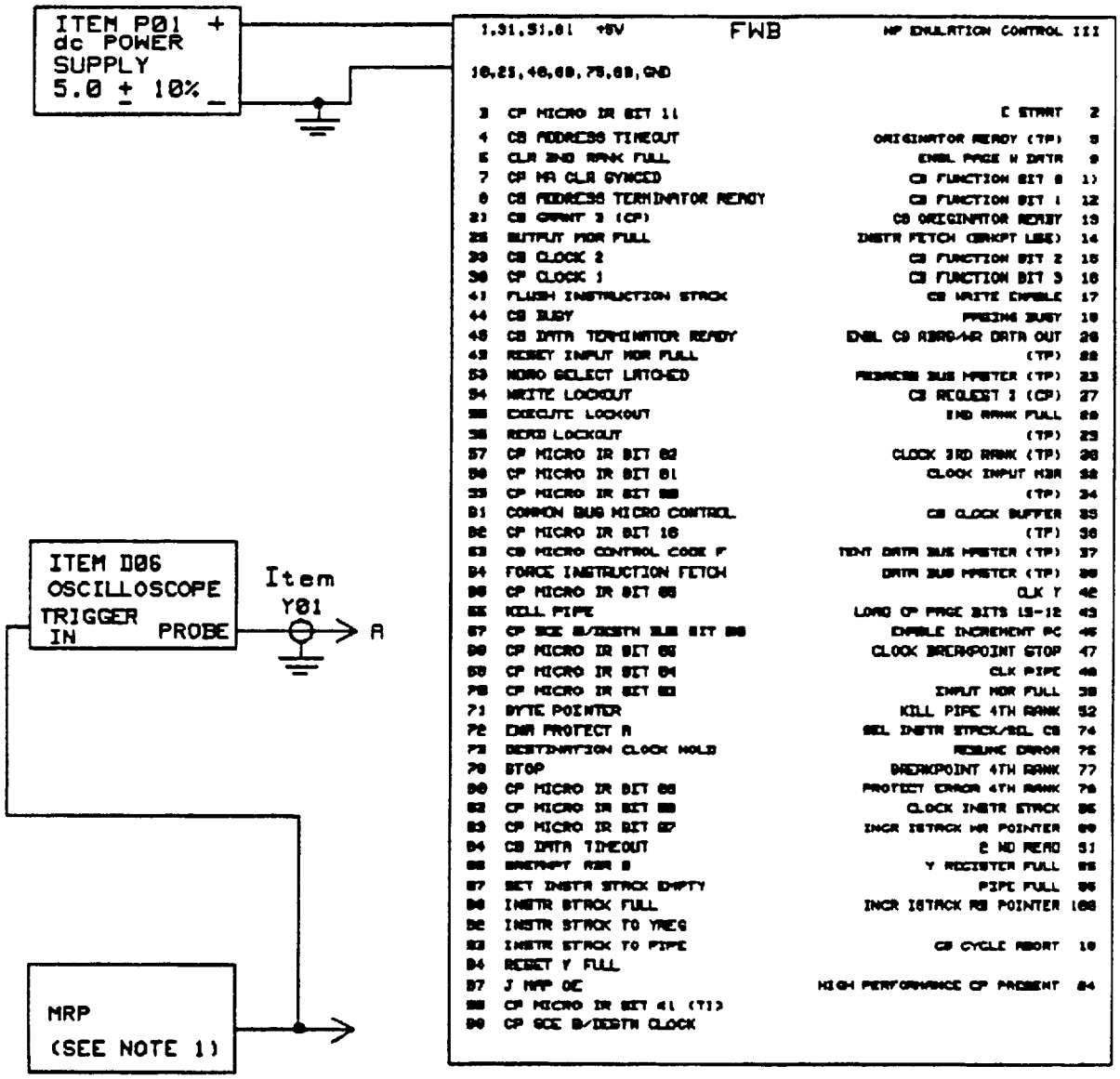


FIGURE 14. Power supply transient amplitude waveform for table XIX.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 15. Power supply transient amplitude test circuit for table XIX.

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